Portable Single-Mirror Schlieren Apparatus

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Design Description

Revision: Oct 4, 2026

This document describes the design of the apparatus, and is maintained authoritatively as Markdown fragments in the project repo on github at https://github.com/fritzm/schlieren/tree/main/docs/design.

Conventions

- The design described here is the baseline. Unless marked otherwise, dimensions, parts, and methods are settled.
- **Provisional** marks a design area, dimension, or method that is still open or evolving, typically pending a physical fit check, measurement, or further design. Provisional areas are rolled up with next steps in §11.
- Measured values, manufacturer specifications, and calculated values are identified as such where the distinction matters; a measured value takes precedence over a catalog value.
- Quantities, vendors, part numbers beyond those needed to define the design, procurement state, and the CAD state of printed parts are recorded in the BOM (`bom/bom.csv`), not here.

1. Project context and design constraints

The apparatus is a portable, single-mirror schlieren system intended initially for several days of vacation/travel use in a location with a large room and a substantial tabletop. It should remain useful afterward as a bench-top laboratory instrument.

The complete system should pack into a standard airline rolling suitcase. Plan around the ordinary 50 lb / 23 kg checked-bag target.

The design is cost-conscious rather than constrained to a fixed dollar ceiling. Commodity 2020 extrusion, Baltic birch plywood, standard fasteners, and printed PLA are preferred where they are adequate. Commercial optomechanics are used where they materially simplify alignment or provide precision that would otherwise be troublesome.

1.1 Fabrication envelope

Near-term 3D printing is PLA only. Critical optical heights and precision datums should be established by metal posts, optical hardware, shim stock, mechanical stops, or adjustment mechanisms rather than relying solely on as-printed PLA dimensions.

PLA softens at about 55–60 °C and creeps under sustained load, so printed parts must not see direct sun or a closed vehicle, and sustained loads on them are kept low: clamp screws are tightened to a light snug, and the preloaded flexures of the slit head are checked for relaxation (§7.7). The LED heatsinks rise only about 14 °C above ambient (§6.3).

The following are out of scope:

- custom machined parts requiring a mill or lathe;
- cutting stainless-steel stock or sheet to custom shape or length.

Use off-the-shelf metal parts, vendor-supplied finished dimensions, plywood, aluminum that can be drilled/tapped with ordinary shop tools, thin foil, or printed PLA. Simple drilling and tapping of the aluminum LED carrier caps and heatsinks is within scope.

A table saw, drill press, and small hand router are available and are expected to be used for accurate working of Baltic birch plywood panels (thanks, Bret!)

2. System overview and physical layout

The apparatus consists of two main physical assemblies:

1. a tabletop optical head; and
2. a separate mirror cell, normally tripod-mounted approximately 3.2 m away.

The tabletop optical head uses two independently yaw-adjustable 2020 aluminum-extrusion rails mounted to a common Baltic-birch front pivot plate. The rails diverge slightly away from the mirror-facing end so that each rail follows one optical chief ray:

- the source rail carries the illumination system directed toward the mirror;
- the imaging rail carries the cutoff and camera system for the return beam.

The source rail contains, in optical order:

1. light source (interchangeable threaded LED/heatsink modules, condenser, and adjustable iris)
2. source-slit flexure head

The slit is the effective optical source. The LED primarily illuminates that slit through the condenser and therefore does not itself require the same precision vertical positioning as the slit.

The imaging rail contains:

1. cutoff carriage at the returned slit-image plane;
2. lens cradle, which supports and aligns the telephoto lens;
3. phone rest, under the iPhone hanging from the lens.

The spherical mirror forms an image of the source slit back near the optical head. Refractive-index gradients in the test region near the mirror deflect portions of this return beam relative to the cutoff, producing schlieren contrast.

Several source-side and imaging-side fixtures deliberately share common mechanics. Three locations use a common Thorlabs TR50/M optical post and printed rail shoe:

1. light source;
2. source-slit;
3. cutoff.

The source slit and the cutoff both rotate about the optical axis on printed fixtures mounted in Thorlabs SM1RC/M SM1 slip rings. The slit uses a flexure head with fine centering and slit-width adjustment (§7); the cutoff uses a carriage that accepts interchangeable cassettes, allowing knife-edge, wire, and color-filter experiments without changing the supporting mechanism (§8).

The mirror cell uses a three-point, approximately 120°-spaced fine-adjustment architecture with compression springs and spherical washer interfaces (§10). The mirror is compliantly supported in the moving plate by six radial silicone pads, which are cast and cured in place between the mirror edge and the inner surface of the plywood aperture.

3. System-level geometry

3.1 Primary optic

- Mirror: Skyoptikst D203F1600
- Diameter: 203 mm
- Focal length: 1600 mm
- Surface: spherical
- Radius of curvature / nominal operating distance: approximately 3200 mm

The source slit and imaging/cutoff system operate near the mirror center of curvature. The test region is near the mirror, where outgoing and returning beams remain close together.

The source slit is the effective optical source. The spherical mirror re-images the slit near the optical head; the returned slit image defines the cutoff plane.

3.2 Source cone

At 3200 mm, the mirror subtends a half-angle of approximately:

$$\theta = \tan^{-1}(101.5/3200) \approx 1.82^\circ$$

The intended illumination cone is approximately ±1.8° to ±2.0°, corresponding to approximately NA = 0.032–0.035 in air. Slight mirror overfill is preferred to underfill.

Useful nominal beam diameters for a ±1.8° cone are:

Distance from source slit	Approx. beam diameter
1 m	63 mm
2 m	127 mm
3.2 m	203 mm

The mirror's approximately f/7.9 focal ratio should not be confused with this source cone. The source operates near 2f, at approximately the mirror center of curvature.

3.3 Source/imager separation

The two rail pivot centers are separated by 95 mm at the mirror-facing front plate. At 3.2 m this gives approximately 1.70° total chief-ray separation.

Each rail points toward the mirror and follows its own chief ray. Sliding a fixture along a rail is therefore predominantly an axial adjustment.

The source slit and cutoff need not occupy exactly the same longitudinal coordinate. Near 2f,

$$\frac{ds_i}{ds_o} \approx -1$$

so small source-slit displacement produces approximately equal and opposite conjugate-image displacement.

A useful starting point is about 30–40 mm total slit/cutoff longitudinal staggering, obtained by placing them approximately ±15–20 mm around nominal 2f. This is an experimental setup adjustment rather than a fabrication-critical dimension.

3.4 Common optical-axis datum

The common optical axis is 72.35 mm above the top surface of the 2020 extrusion.

For optical-head CAD:

- take the rail top surface as z = 0;
- do not reference optical height to the tabletop or support feet;
- small whole-frame pitch/roll errors can be corrected at the mirror.

The optical head uses three coordinate frames with common senses:

- +y forward, toward the mirror;
- +x to the right when looking down from above with the mirror ahead;
- +z up, with the rail top at z = 0.

The frames are right-handed and differ only in where their y axes lie:

- pivot frame: y along the centerline of the front pivot plate, with the origin at the midpoint of its aft edge (§4.2);
- source rail frame: y along the source rail centerline;
- imaging rail frame: y along the imaging rail centerline.

The imaging rail is the right rail, at positive x in the pivot frame, and the source rail is the left. The phone extends well to the right of its camera axis when sighting along that axis toward the mirror (§9.1), so this places its overhang on the outer side of the imaging rail, clear of the source rail and its fixtures.

Each rail frame yaws with its rail, about 0.85° from the pivot frame at nominal (§4.2). Its origin is on the rail top at the rail's pivot axis; a fixture is described with its own rail station at y = 0. In a rail frame the optical axis is at x = 0, z = 72.35 mm, and pitch, yaw, and roll are rotations about x, z, and y respectively. A rotating fixture may define a local frame, stated where it is used.

The common post stack is:

- 0.010 in / 0.254 mm metal datum shim;
- 50 mm Thorlabs TR50/M post;
- 22.1 mm post-top-to-optical-axis offset for the SM1 optical holders (Thorlabs SMR1/M: 0.870 in / 22.098 mm exact);
- total nominal height: 72.35 mm (72.352 mm with the exact SMR1/M offset; the datum is 72.35 mm).

The datum discs are McMaster 2895T62, 3/4 in OD × 0.010 in ±0.0005 in thick, full-hard 18-8 stainless steel, minimum Rockwell C40. Three discs are used as post datum shims, one under each common post fixture. The discs provide a consistent metal seating surface for the posts across the central slot in the 2020 rail.

4. Tabletop optical frame

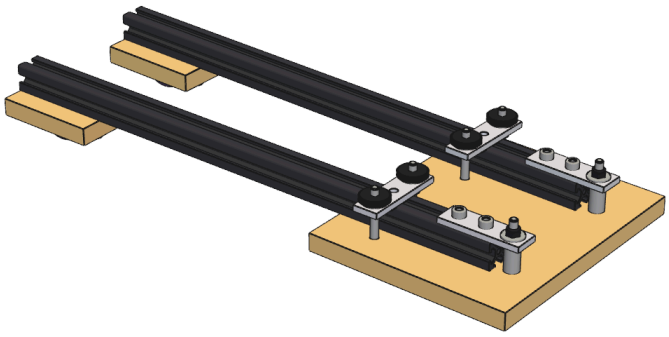


Fig. 1. Tabletop frame: rails, front pivot plate with pivot lugs and yaw straps, and rear foot blocks

The model is `src/schlieren/parts/frame.py`.

4.1 Rails

- IXGNIJ black-anodized metric 2020 extrusion, Amazon B08Y8N7FD1
- 20 × 20 × 400 mm
- two rails used
- nominal 6 mm European-style slot

There is no rear crossrail.

Measured slot geometry of the extrusion:

Feature	Measured value
Slot mouth width	0.258 in / 6.55 mm
Internal cavity width	0.436 in / 11.07 mm
Slot depth	0.254 in / 6.45 mm
Lip thickness	0.065 in / 1.65 mm

These dimensions suit M5 hardware for 20-series / nominal slot-6 extrusion. Ordinary attachments use M5 × 0.8 socket-head hardware with flat washers in standard metric lengths and the roll-in/drop-in M5 T-nuts (Amazon B0DP6JBX4Y).

4.2 Front pivot plate

The front pivot plate is a 180 × 150 × 12.7 mm Baltic-birch rectangle with no structural cutouts. The 180 mm dimension is transverse to the rails and the 150 mm dimension is fore/aft.

Plate coordinates are the pivot frame (§3.4): the origin is at the midpoint of the aft edge, +y runs toward the mirror along the plate centerline, and +x runs to the right seen from above with the mirror ahead. The mirror-facing edge is at y = 150 mm, and the left rail is the one at negative x. The pivot line is 25 mm behind the mirror-facing edge, giving pivot centers at:

- left pivot: (-47.50, 125.00) mm;
- right pivot: (+47.50, 125.00) mm;

- total pivot separation: 95 mm.

The nominal rail half-angle implied by the 95 mm pivot separation and approximately 3.2 m mirror distance is about 0.850°, or about 1.70° total chief-ray separation.

Each yaw-clamp station is 90 mm aft of its pivot along the nominal rail axis, giving yaw-strap centers at approximately:

- left strap center: (-48.84, 35.01) mm;
- right strap center: (+48.84, 35.01) mm.

Each fixed yaw strap is perpendicular to its rail at the nominal rail angle. The two outer holes of each 3-hole joining plate give fixed yaw-bolt centers at approximately:

- left outer: (-68.83, 35.31) mm;
- left inner: (-28.84, 34.71) mm;
- right inner: (+28.84, 34.71) mm;
- right outer: (+68.83, 35.31) mm.

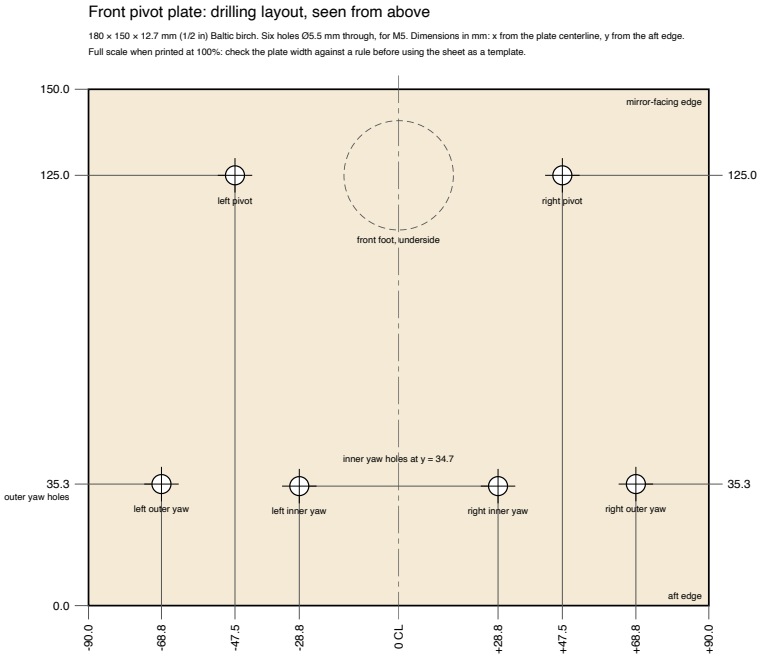


Fig. 2. Front pivot plate drilling layout, seen from above, in plate coordinates

The four fixed yaw bolts pass through approximately 5.5 mm round clearance holes; the plywood is not slotted. The straps stay fixed to the pivot plate while the 20 mm rails rotate beneath them. With 40 mm yaw-bolt spacing, the rail has approximately 7.5 mm nominal edge-to-M5-shank clearance per side and retains about 2.8 mm clearance at ±3° rail yaw from nominal for the 90 mm pivot-to-clamp distance. Rail-to-bolt contact does not occur until roughly ±4.7° from nominal, so the fixed-hole geometry provides ample adjustment range.

A front Sorbothane support foot is centered at approximately (0, 125) mm, directly between the two rail pivots, keeping the support point on the pivot line and clear of the pivot and yaw hardware.

1/2 in Baltic-birch sheet stock is shared across the tabletop frame and mirror cell: it supplies the front pivot plate, the two rear rail-support/foot blocks, the mirror-cell cell-adjuster plate, and the mirror-cell base plate.

4.3 Pivot and yaw hardware

Pivot and yaw hardware use commercially available 2020 three-hole joining plates:

- MOPFOL 2020 3-hole joining plate, Amazon B0G8DCS7VP
- approximately 60 × 18 × 4 mm
- 20 mm hole pitch
- four in all: two as pivot lugs and two as yaw straps.

Pivot stack

Per rail, from the bottom up:

socket head → oversized washer → plywood pivot plate → 20 mm spacer → 4 mm joining plate → oversized washer → M5 nyloc

The socket heads are beneath the plywood and the nylocs on top, matching the yaw-lock screws.

Hardware:

- McMaster 92290A265, M5 × 50 partially threaded 316 SS socket-head screw
- McMaster 92871A091, 20 mm × 10 mm OD × 5.3 mm ID stainless spacer
- McMaster 91116A350, M5 oversized washer, 15 mm OD
- McMaster 93625A225, M5 nylon-insert locknut

The pivot screws pass through approximately 5.5 mm clearance holes in the plywood.

Yaw lock

One 3-hole joining plate spans transversely across each rail. The two outer holes are used, giving 40 mm fixed bolt-center spacing. The strap is fixed to the plywood by two round-hole through-bolts, one on each side of the rail; the rail itself rotates beneath the strap. The plywood has no slotted yaw holes.

Each yaw screw stack, from the bottom up:

socket head → oversized washer → plywood pivot plate → (alongside the rail) → 4 mm yaw strap → oversized washer → M5 thumb nut

The thumb nuts are on top of the straps, where they are accessible with the frame standing on its feet.

A strip of McMaster 9852N37 adhesive-backed 1/32 in (0.79 mm) EPDM on the underside of each strap, where it bears on the rail top, adds friction to the yaw lock. The strip spans the strap width and stays between the bolt holes (about 30 mm long, clear of the holes at ±20 mm).

Each strap center is 90 mm aft of its rail pivot along the nominal rail axis, with the strap perpendicular to the nominal rail direction (§4.2). Yaw is adjusted and locked at the front pivot plate by loosening the two thumb nuts, rotating the rail beneath the fixed strap, and retightening. The bolt/rail clearance supports at least approximately ±3° adjustment about nominal with useful remaining clearance.

Hardware:

- McMaster 92290A265, M5 × 50 partially threaded 316 SS socket-head screws — same fastener as the rail pivots
- McMaster 91116A350 oversized M5 washers
- McMaster 92815A202 M5 low-profile knurled thumb nuts
- McMaster 9852N37 1/32 in EPDM friction strips, cut from the same sheet as the cassette clamp-bar pads

The yaw screws pass up alongside the 20 mm rail, not through it, and span the approximately 1/2 in (12.7 mm) Baltic-birch pivot plate, the full 20 mm rail height, the EPDM strip, the 4 mm yaw strap, washers, and the thumb-nut engagement. The M5 × 50 yaw-clamp screw deliberately reuses the pivot-screw part rather than introducing a separate low-profile screw.

4.4 Rear support and feet

Each rail has a removable 75 × 50 × 12.7 mm (3 × 2 × 1/2 in nominal) Baltic-birch rear foot block, cut from the same 1/2 in sheet stock used elsewhere in the frame and mirror cell. The 75 mm dimension runs along the rail and the 20 mm rail is centered across the 50 mm block width.

Each block attaches to the rail bottom slot with two M5 socket-head screws, two flat washers under each head, and roll-in/drop-in M5 T-nuts. The two screw holes lie on the rail centerline and are 50 mm apart, symmetrically placed about the block center; with the 75 mm block length, each screw center is 12.5 mm from its nearest end. The screws and washers are M5 from the general M5 assortment; the screws are 20 mm long, through approximately 5.5 mm clearance holes in the plywood.

The two washers set the screw's depth in the slot. With two nominal 1.0 mm washers and the 12.7 mm block, a 20 mm screw enters the rail slot by about 5.3 mm, about 1.15 mm short of the 6.45 mm slot floor. A single washer leaves only about 0.15 mm, so the screw bottoms before the block clamps if the plywood runs thin; a 16 mm screw reaches only about 0.65 mm past the slot lip, too little to engage the T-nut; no standard length lies between.

Each rear-foot screw stack, from the bottom up:

socket head → two flat washers → plywood foot block → M5 T-nut in the rail bottom slot

A Sorbothane foot is centered on the underside of each block, directly beneath the rail centerline and midway between the two attachment screws. The 1-1/4 in / 31.75 mm diameter foot therefore does not obscure either attachment screw or washer, allowing the complete plywood/foot assembly to be removed from and reinstalled on the rail without disturbing the Sorbothane foot. The blocks and feet support the rear of each rail but do not provide yaw lock.

The complete frame is therefore supported on the table/bench top at three points:

- one centered foot beneath the front pivot plate;
- one foot beneath each rear support block.

Feet: McMaster 8215K2, 1-1/4 in OD × 5/8 in high, 30 OO Sorbothane, with a stiffer McMaster 8215K6 (same size) as an alternative depending on vibration isolation needs. Moderate static compression of the front foot is acceptable; the resulting small overall pitch is corrected during optical alignment.

5. Common rail-mounted post support

Three fixtures use the common post/shoe architecture:

1. light source (§6);
2. source-slit flexure head (§7);
3. cutoff carriage (§8).

5.1 Post

- Thorlabs TR50/M
- 12.7 mm diameter
- 50 mm long

5.2 Optical-holder allocation

- light source: 1 × SMR1/M (§6.1)
- slit head: 1 × SM1RC/M, clamping the printed spigot of the slit-head adapter (§7.6)
- cutoff carriage: 1 × SM1RC/M, clamping the printed spigot of the base plate (§8.1)

The source/slit/cutoff holders preserve the 22.1 mm post-top-to-axis dimension.

5.3 Common rail shoe

The common shoe is a printed PLA U-shaped saddle that:

- straddles the 2020 extrusion;
- locks through the rail side channels with M5 screws into the measured slot geometry (§4.1), using the 20-series slot-6 T-nut hardware, with a flat washer under each screw head; the screws and washers are from the general M5 assortment;
- leaves the rail top available as the datum surface;
- carries a vertical 12.7 mm through-bore for the TR50/M post.

The post passes through the printed shoe and bottoms on a McMaster 2895T62 0.010 in stainless datum disc resting on the rail top. The printed PLA does not establish the precision optical height.

The post bore incorporates a single-split clamping collar so that the post can be locked against rotational movement while remaining fully seated on the metal datum:

- one axial split;
- one transverse clamp screw;
- sufficient axial engagement to resist rocking and rotation;
- rounded stress-relief termination at the bottom of the split;
- generous fillets where the collar merges into the shoe body;
- free-fitting 12.7 mm post bore before clamping.

The clamp provides rotational and lateral retention; post height is still established solely by the post bottoming on the metal datum disc. The model, with its PLA fit allowances, is `src/schlieren/parts/rail_shoe.py`.

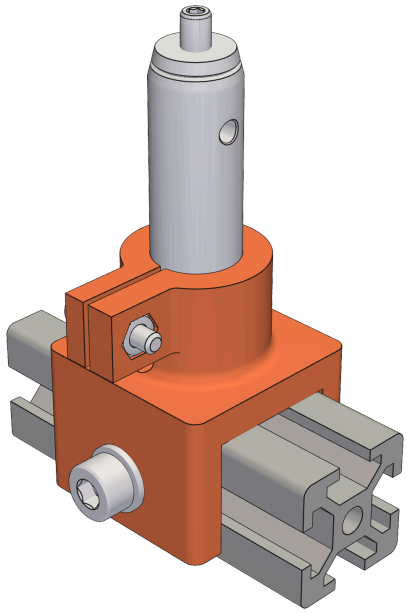


Fig. 3. Common rail shoe on rail with TR50/M post

Common printed split-clamp hardware

Every printed split clamp in the apparatus uses the same hardware:

M3 socket-head screw → clearance through one ear → captured M3 hex nut in the opposite ear.

- McMaster 91292A114 — M3 × 0.5 × 12 mm fully threaded 18-8 stainless socket-head screw; 5.5 mm head diameter, 3 mm head height, 2.5 mm hex drive. The same screw serves the carriage keeper frame (§8.4) and the slit head (§7).
- McMaster 91828A211 — full-height M3 × 0.5 18-8 stainless hex nut; 5.5 mm across flats × 2.4 mm high, DIN 934. The same nut serves the keeper frame, the cassette clamp bars (§8.7), and the slit head.
- No washer, nyloc, or threaded insert.

Ear thickness and the approximately 1–2 mm relaxed split gap are designed around the 12 mm screw length.

6. Light source

The light source is a single assembly on one post and common rail shoe (§5). A threaded holder carries one of two interchangeable threaded LED modules on its rear face, and a condensing lens and iris on its forward face. The condenser images the LED emitter onto the source slit (§7), and the iris sets the angular size of the illumination cone (§3.2).

`src/schlieren/parts/light_source.py` models the stack-up; `uv run light-source` prints the focus ranges.

6.1 Assembly and stack order

From the LED to the slit:

1. threaded LED module (§6.2), with Thorlabs SM1CP2M cap threaded into the LED-side face of the SMR1/M;
2. Thorlabs SMR1/M mount (accepts the post);
3. Thorlabs SM1V05, with its externally threaded sleeve entering the slit-side face of the SMR1/M and locked with the included SM1NT locking ring. A Thorlabs ACL2520U-A condensing lens sits in the SM1V05 body with its plano face on the internal seat (toward the LED) and its convex face toward the open end;
4. Thorlabs SM1L03 spacer ring, threaded into the SM1V05 open end;
5. Thorlabs SM1D12 adjustable iris, threaded into the SM1L03.

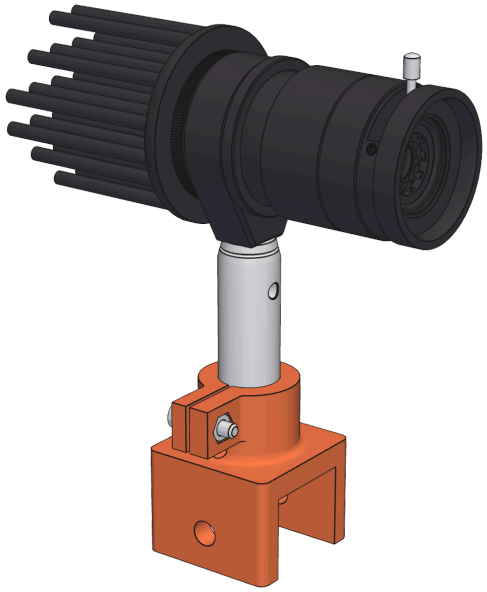


Fig. 4. Light source on its post and rail shoe: heatsink and LED module, SMR1/M, SM1V05, SM1L03, and iris

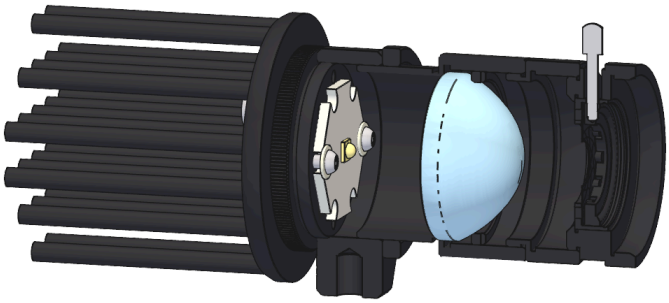


Fig. 5. Light source cutaway, showing internal position of LED module and condensing lens

The ACL2520U-A convex vertex sits about 0.7 mm inside the SM1V05 open end, so the SM1L03 spacer is required ahead of the iris; the SM1D12 interferes with the lens if threaded directly into the SM1V05.

The cap and the SM1V05 sleeve share the SMR1/M thread. The cap threads in until its knurled flange seats on the SMR1/M face, so the LED position is fixed, and the SM1V05 engagement is the focus adjustment (§6.7). Engagement runs from the Thorlabs minimum of 0.110 in / 2.79 mm to about 5.7–5.8 mm, where the sleeve end comes within 0.3 mm of the LED board. Turning the SM1V05 moves the lens and iris together relative to the LED; the resulting few-millimeter iris shift is negligible against the 150 mm iris-to-slit spacing.

6.2 LED modules

There are two interchangeable LED modules, each a self-contained threaded assembly. The LEDs chosen are on industry standard 20mm "star" MCPCB carriers. Each carrier is mounted on an identical threaded SM1 threaded cap, so the SM1 thread and cap flange are the interchange standard.

Module	Emitter	Board	Initial operating current (mA)
Green	OSRAM OSLON SSL 120, 530 nm nominal	NewEnergy star board, LST1-01F06-GRN1-00	~350
White	Nichia 519A R9080, 5000 K	Convoy 20 mm star/DTP carrier	~500–700

Module stack, from the rear toward the condenser:

1. Alpha CN40-40B heatsink (§6.3), bolted to the rear face of the cap with thermal compound between;
2. Thorlabs SM1CP2M externally SM1-threaded end cap (0.210 in / 5.33 mm solid aluminum, 0.100 in / 2.54 mm thread, Ø1.200 in / 30.48 mm knurled flange);
3. LED star board on the front face of the cap (§6.4).

Interfaces:

- The cap flange is the repeatable axial stop on the SMR1/M face; the module's rotational position at the stop does not matter.
- The SM1 thread centers the emitter on the condenser axis; no vertical calibration is required.
- Heatsink fastening: three M3 × 6 mm socket-head screws (**provisional** length) through the CN40 base, their heads in open cells between four pins, into M3 holes tapped through the cap. Positions are in §6.4.
- Leads drop past the star board's flat edges through two Ø2.0 mm holes in the cap and the heatsink base, 11.0 mm from the axis (inside the thread root), and out between the pin columns. **Provisional:** the hole size against the lead insulation (§6.4).
- Fabrication is drilling and tapping of purchased aluminum parts only. Every hole in the cap and the heatsink base is a through hole, so plain plug taps are enough.

Clearances:

- The heatsink's lowest point is 2.1 mm above the TR50/M post top at any rotation (22.10 mm axis offset minus 20 mm maximum radius).
- The heatsink front face is 2.79 mm (0.110 in) behind the SMR1/M face (the cap flange thickness).
- The SMR1/M thread bore leaves about 3 mm of radial room around a 20 mm board for leads.

6.3 Common heatsink

- Alpha CN40-40B natural-convection pin-fin heatsink, one per module
- Ø40.0 mm (+0/−0.5) × 40 mm overall, 3.0 mm flat base, Ø2.8 mm pins on a 6.9 mm square grid
- 6063 aluminum, black anodized; 24.6 g
- manufacturer natural-convection rating 6.9 °C/W

At about 2 W (white module at 700 mA) the expected rise is roughly 14 °C above ambient; the green module dissipates about 1.1 W. The rating is the manufacturer's pins-up test; the module runs with pins horizontal, which pin fins tolerate well. **Provisional:** bench-test the temperature rise at 700 mA before relying on it.

The heatsink is round so that it clears the post at every rotation while the module is threaded in (§6.2); a larger or non-round heatsink would sweep its corners below the post top during threading.

6.4 LED board mounting

Measured star-board thicknesses (calipers):

- green NewEnergy star: 0.0625 in / 1.59 mm (the datasheet lists 0.047 or 0.067 in; the measured value governs)
- white Convoy 20 mm star: 0.0590 in / 1.50 mm

Both boards are about 20 mm across (NewEnergy 0.783 in / 19.9 mm, catalog).

Fastening standard:

- M2 × 0.4
- McMaster 92095A452, M2 × 5 mm screws
- tapped directly into the front (threaded) face of the SM1CP2M cap, in through holes
- 1.6 mm tap drill, drilled through
- M2 × 0.4 HSS plug tap; a through hole needs no bottoming tap

The 5 mm screws are shorter than the 5.33 mm cap, so they stop short of the heatsink face.

Use the cap's center-drill mark as the layout datum for the board hole pattern. Transfer each carrier board's hole geometry from the physical board rather than assuming it from the other module. Slightly oversized board holes permit final centering of the emitter on the cap axis: a 0.3 mm emitter offset moves the slit image about 2 mm, so center each board by watching its image on the slit blades before final tightening.

A thin insulating washer goes under each screw head, keeping the metal head off exposed board contacts and spreading its clamp over the board beside the notch:

- McMaster 91545A410
- M2 lubricant-filled nylon flat washer

A thin layer of thermal compound is used between board and cap.

Cap and heatsink hole layout

All holes in the cap and in the heatsink base are **through holes**, drilled from one face. Plain plug taps (M3 × 0.5, M2 × 0.4) cut full threads through them; no bottoming tap is required. No screw reaches the far face of the cap: the M2 screws are 5 mm under the head and the M3 screws engage about 3 mm past the 3.0 mm heatsink base.

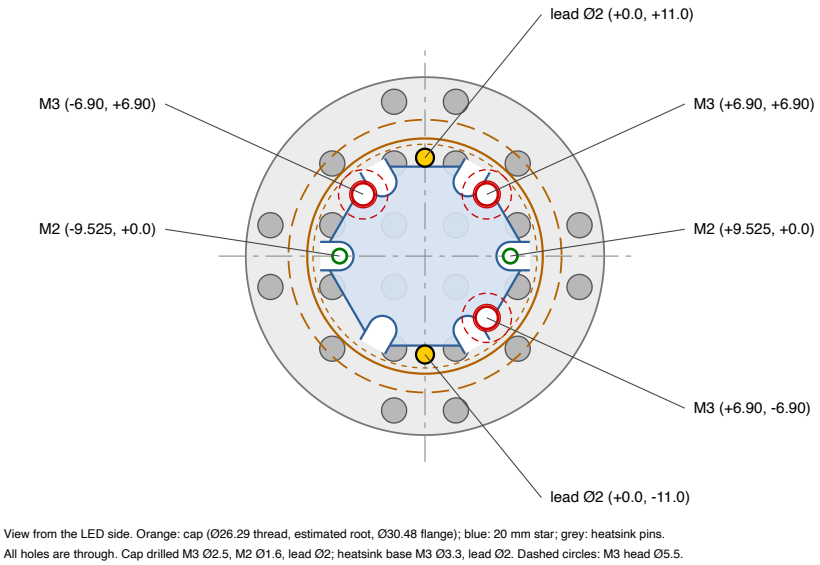


Fig. 6. Cap and heatsink hole layout, seen from the LED side, in the frame of the heatsink pin lattice

Coordinates are in millimeters from the optical axis, +x to the right and +y up as seen from the LED side, in the frame of the pin lattice; the cap and the heatsink are drilled to the same coordinates:

Hole	Position	In the cap	In the heatsink base
M3, three	(+6.90, +6.90), (−6.90, +6.90), (+6.90, −6.90)	Ø2.5 tap drill, M3 × 0.5	Ø3.3 clearance
M2, two	(±9.525, 0)	Ø1.6 tap drill, M2 × 0.4	none
Lead, two	(0, ±11.0)	Ø2.0	Ø2.0

- M3 holes: each is the center of a cell of the pin lattice, 4.88 mm from each of four pins, so the Ø5.5 mm socket head lands between the pins with 0.73 mm to spare and the key reaches it. Three screws hold the heatsink without a rocking axis; the fourth cell is free but unused. Roughly 70% of each hole's area lies under the star, a small loss of its thermal contact.
- M2 holes: in the two opposite corner notches of the star on the x axis (the 0.375 in notch center radius). They are covered by the solid heatsink base, so they do not meet the pins.
- Lead holes: on the axis through the midpoints of two opposite flat edges of the star, tangent to each edge at 11.0 mm radius. Each is 1.1 mm from the nearest pins, in the 4.1 mm channel between pin columns.
- The smallest wall between holes in the cap, measured to the thread major radius, is 4.9 mm. The lead holes' outer edges, at 12.0 mm radius, leave about 0.5 mm to the estimated thread root.

Provisional: the lead hole size against the insulation of the leads (a Ø2.0 hole leaves no room for thicker wire, since the outer edge is limited by the thread root); the cap thread root radius (estimated at 12.5 mm, to be measured on a cap); and the Convoy star's notch geometry, which is taken from the NewEnergy drawing and must be checked against the physical board.

Full-scale drilling templates for both parts, to print at 100% and stick on, come from `uv run light-source --templates`: the cap template goes on the LED-side face with its crosshair on the center-drill mark, and the heatsink template on the flat base face, aligned to the pin rows. Both are viewed from the LED side, so they share one set of coordinates.

6.5 Power and wiring

Power supply:

- ALIENTEK DP100
- use DC constant-current operation rather than PWM to avoid rolling-shutter banding

Connector standard:

- Micro-Fit 3.0
- 2-pin
- 18 AWG factory pigtails

Connector hardware:

- 2 × Molex 214758-1021, male-to-pigtail, 150 mm — module side
- 1 × Molex 214756-1023, female-to-pigtail, 600 mm — supply side
- 1 × Pomona 1825-02 red/black 4 mm banana-plug pair

The supply side should present recessed/socket contacts.

6.6 Condenser optics

The condenser images the LED emitter onto the source slit (about 6.6× magnification). Each slit point then sees the iris uniformly filled at the LED radiance, which gives a uniformly illuminated mirror. The LED image should cover the illuminated slit length; the domes enlarge the apparent emitters by about 1.4–1.5×.

Defocus in either direction underfills the ±1.8° cone: iris-edge rays then trace back beyond the emitter edge, darkening the mirror edge and shortening the uniformly lit slit length. With the LED at the condenser focus (12 mm, collimated) the emitter cannot fill the cone even at the slit center.

Using ACL2520U-A catalog values (EFL 20.1 mm, BFL 12.0 mm; object-side principal plane about 8.1 mm inside the plano face) and about 153 mm from the convex vertex to the slit, the optimum emitter-to-plano-face gap is about 15 mm. The optimum moves only about ±0.7 mm for ±30 mm of slit position.

6.7 Focus and module interchange

Setup values (not hard CAD dimensions):

- emitter-to-plano-face gap approximately 15 mm, measured to the emitting surface, not the dome top
- useful gap range roughly 13–17 mm
- condenser/iris-to-slit spacing approximately 150 mm
- initial iris opening approximately 9–10.5 mm (about ±1.7–2.0° at 150 mm)

Set focus by imaging the LED onto the closed slit blades and turning the SM1V05 until the emitter image is sharp and centered, then lock the SM1NT and open the slit.

Calculated emitter heights above the cap face, using the measured board thicknesses, about 0.07 mm of solder, and the apparent emitter heights through the LED domes (calculated from the manufacturer outline drawings: OSLON about 0.45 mm, 519A about 0.75 mm), with the gap range each module reaches over the SM1V05 engagement range:

Module	Emitter height above cap face	Emitter-to-plano gap range	SM1V05 engagement at 15 mm gap
Green	about 2.1 mm	about 13.2–16.2 mm	about 3.9 mm
White	about 2.3 mm	about 12.9–16.0 mm	about 3.7 mm

The two modules focus about 0.2 mm apart in engagement, about one-third of a turn of the SM1 thread. An index mark per module on the SM1V05 is optional.

Focus margin about the 15.04 mm optimum gap is about 1.1 mm (long-gap side) and 1.8 mm (short-gap side) for the green module, and only about 0.93 mm and 2.1 mm for the white module. The white module's long-gap travel is limited by the SM1V05 minimum engagement. **Provisional:** the white module's reduced margin is to be confirmed on the build. If it cannot reach a sharp slit image, space its cap flange off the SMR1/M face with a thin shim (about 0.1 mm restores 1.0 mm of margin). The shim moves the whole module away from the lens and keeps the flange as the axial stop; a shim under the star board would instead shorten the gap and make the shortfall worse.

Module interchange:

1. disconnect the Micro-Fit lead;
2. unscrew the module (cap and heatsink together) from the SMR1/M, about four turns;
3. screw in the alternate module until its flange seats;
4. touch up focus on the closed slit blades if needed.

In transport, the condenser/iris assembly is protected from dust and scratches by a Thorlabs SM1CP1 internally threaded end cap and an SM1EC2 snap-on dust cap.

7. Source slit

The source slit is a flexure slit head: one flat-printed PLA part carrying both blades, with fine centering perpendicular to the slit and fine slit-width adjustment, mounted for continuous rotation about the optical axis in an SM1RC/M on the common post and rail shoe (§5).

The model is `src/schlieren/parts/slit_head.py` (`uv run slit-head`, with `--show` for the viewer assembly).

The slit needs only about ±2 mm of centering travel, one adjustment axis, and discrete orientations, so it does not use the cutoff's carriage, plungers, guide rails, keeper frame, datum pins, or cassettes (§8).

The stage-rotation figures below come from a first-order flexure model, not FEA.

7.1 Blades and slit widths

The slit uses two Stanley 11-515 backed single-edge scraper blades.

Measured physical blade dimensions:

- length: 1.535 in / 38.99 mm
- width: 0.768 in / 19.51 mm
- thickness at the sharp/flat portion: 0.010 in / 0.254 mm
- thickness at the folded spine: 0.040 in / 1.016 mm

Expected operating slit widths are 0.10–0.30 mm, starting around 0.15–0.20 mm.

Each blade lies flat on its carrier's blade seat, cutting edge toward the slit and overhanging the carrier edge by about 1.5 mm. No blade pockets, holes, or spine compensation are used; a slight inclination of the blade toward its folded spine is acceptable, since the optically important geometry is the gap and parallelism of the two cutting edges.

7.2 Architecture

The head is a three-body monolithic flexure:

1. frame — fixed; carries the spigot adapter, the FAS100 #1 insert, and the spring seat;
2. platform — centering stage on a parallelogram flexure from the frame; carries the datum blade and the FAS100 #2 insert;
3. width stage — on a nested parallelogram flexure from the platform; carries the movable blade.

Both parallelograms translate perpendicular to the slit. Their parasitic motion (about δ^2/L) lies along the slit, where it is harmless.

Local frame, at rotation 0: slit horizontal, adjustment axis vertical, both FAS100 knobs up. The blade-seat plane (the carrier front faces) is the front of the head; every other front face stands back 1 mm so the overhanging blades cannot rub moving or fixed parts.

The head's front faces the light source, so the LED image on the blades is in plain view for centering and focus (§6.7). The spigot adapter, slip ring, and post are on the mirror side of the blades. In this section "front" and "back" refer to the head, not to the rail direction.

Envelope: about 80 × 88 × 13 mm, excluding knobs and adapter.

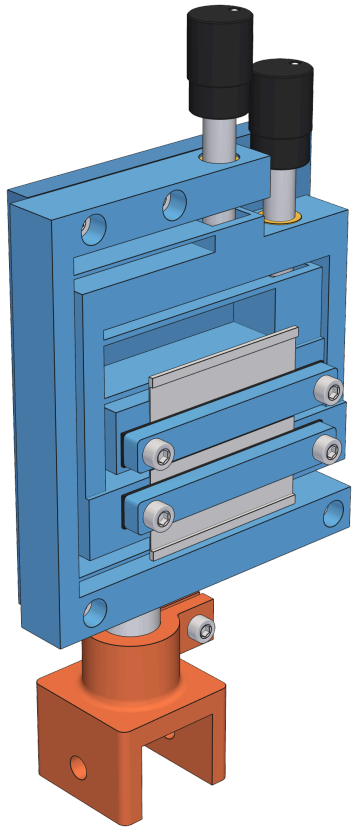


Fig. 7. Flexure slit head on its post and rail shoe, seen from the light-source side: blades and clamp bars, FAS100 centering and width adjusters

Stage rotation in the slit plane would skew the slit against the cutoff (centering) or open it in a taper (width). Under load each flexure blade bends into an S with its inflection at mid-span, so only a force through the blades' mid-span needs no axial reaction; the layout keeps the varying drive forces from twisting the stages:

- FAS100 #1 and the preload spring are coaxial and act on the platform's flexure connectors, so their forces cancel and only the small centering-flexure force is unbalanced;
- the two centering blades are at the top and bottom of the platform, about 63 mm apart;
- the two width-stage blades are 12 mm apart, and FAS100 #2 pushes on the width-stage connector.

Closely spaced blades or a spring offset from the screw would not achieve this: a compact layout with 5 mm blade spacing and the spring 23 mm from the screw gives a constant ~1° platform pre-twist and about 8 mrad of rotation over the centering travel.

7.3 Centering stage

- travel: ±2.0 mm about the as-printed position;
- drive: Thorlabs FAS100 1/4"-80 adjuster, 0.3175 mm per turn (10 µm per about 11° of knob), in a McMaster 98625A950 brass insert in the frame top bar;
- preload: McMaster 2006N292 spring, coaxial with FAS100 #1, 16.5 mm long at mid travel (9 mm compression), about 4.4–6.9 N over the travel; located by printed guide pins inside the coil in pockets in the frame bottom bar and the lower platform connector;
- flexure: two 0.8 mm PLA blades, 35.5 mm long, 12 mm deep; about 0.38% peak strain at full travel;
- estimated slit rotation: about 0.05 mrad over the full ±2 mm (0.5 µm across a 10 mm lit slit length).

Measured spring OD: 0.272 in / 6.91 mm (calipers). Spring pockets are Ø7.91 mm (1.0 mm diametral clearance).

Spring installation: the pockets are closed on all sides except the 3.5 mm flexure gap, and the 16.5 mm cavity is shorter than the 25.5 mm free length, so the spring has no clear insertion path. It is installed by compressing it fully and working it into the pockets by hand. If that proves unreliable, the fallback is a bottom access bore through the frame bottom bar closed by a screwed cap carrying the lower guide pin (not modeled).

Sensitivity: the iris is about 150 mm before the slit, so moving the slit by d turns the illumination cone by d/150; at the mirror the beam moves about 21 × d. The ±2 mm travel therefore steers the beam about ±42 mm at the mirror, which covers printed-part and shoe tolerances plus deliberate aiming. For a horizontal slit this is the only vertical aiming adjustment at the source.

7.4 Width stage

- drive: a second FAS100 in a 98625A950 insert in the platform top bar; pushing closes the slit;
- preload: the stage's own flexures, printed so that FAS100 #2 always deflects them 0.8–1.6 mm (about 0.8–1.6 N, comfortably above the stage's own weight in any orientation);
- usable width adjustment: 0.8 mm;
- flexure: two 1.2 mm PLA blades, 45 mm long, 12 mm apart; about 0.29% peak strain;
- estimated taper: about 0.6 µm across a 10 mm lit slit length per 0.1 mm of width change (about 4.6 µm over the full 0.8 mm).

Only the movable blade moves, so a width change shifts the slit center by half the change. For a 0.1 mm change that is 0.05 mm, which moves the beam about 1 mm at the mirror and needs no recentering of the slit (the cutoff is readjusted for any width change in any case).

Slit setup:

1. with both clamp bars loose, set the width stage to mid travel with FAS100 #2;
2. clamp the datum blade on the platform carrier;
3. set the movable blade against a true-metric feeler gauge between the cutting edges, pressing gently along its length, and clamp it progressively;
4. remove the gauge; thereafter set width with FAS100 #2 from that reference (0.15 mm is about half a turn). Re-check parallelism with the gauge after large width changes.

7.5 Blade clamping

Each blade is pinched against its carrier by an identical removable printed PLA clamp bar (54 × 7 × 4 mm) faced with McMaster 9852N37 1/32 in EPDM. The bar bears on the flat portion of the blade about 3 mm back from the cutting edge, leaving the slit cone clear, and is fastened by two M3 × 12 mm socket-head screws just beyond the blade ends into M3 hex nuts captured in pockets opening from the carrier back. The common McMaster 91292A114 screw and 91828A211 nut are used; four of each.

7.6 Rotation mount and clearances

A separate printed spigot adapter is bolted to the frame back with four M3 × 12 mm socket-head screws, counterbored from the head front into M3 nuts captured in the adapter back. The adapter is a flat plate, held 1.0 mm off the moving stages by two McMaster 93475A210 M3 flat washers (0.4–0.6 mm each) stacked on each screw between frame and plate. It carries a hollow Ø30.48 mm (SM1 tube OD) spigot with a Ø24 mm light bore, clamped by the SM1RC/M split ring (Ø30.6 mm bore, 10.16 mm thick, M4 locking screw across the split).

A Ø34 mm shoulder on the spigot is the ring's axial stop: the ring seats against it, which places the adapter plate 3.5 mm in front of the ring face and 2.2 mm clear of the post at every rotation. Do not clamp the ring away from the shoulder.

The SM1RC/M keeps the 22.1 mm post-top-to-axis dimension (§5.2). Loosening the ring rotates the whole head about the optical axis:

- rotation 0: horizontal slit, vertical centering, knobs up;
- ±90°: vertical slit, horizontal centering, knobs sideways;
- small rotations: slit-to-cutoff parallelism trim.

Calculated clearances over −110° to +110°: at least 2.2 mm to the post body, 4.0 mm to the rail shoe (at about ±50°, as the corners pass). The TR50/M M4 stud reaches up into the ring's tapped hole and stops about 1.7 mm short of the spigot at its 5.2 mm drawing maximum; this gap is set by the Thorlabs parts, not by this design. Beyond about ±125° the adapter corners and knobs reach the shoe.

7.7 Fabrication and fit

Printed parts: the flexure head, the spigot adapter, and two clamp bars.

- Head: print back face down, so the blade seats on both carriers are the topmost layer and coplanar (ironing recommended); the flexure blades are then vertical walls loaded in the layer plane. Keep perimeter seams off the four flexure blades.
- Adapter: print front face down, spigot up; no supports.
- Clamp bars: print flat, EPDM face down.
- The insert bores are pilots, finished by a light hand ream with a 5/16 in twist drill to a press fit (§8.3). The FAS100 #1 insert is pressed in from the inside with its flange on the bar's inner face, so the spring load seats the flange against the bar instead of pushing the insert out; the FAS100 #2 flange is outboard, since the platform top bar leaves no room to press it in from the inside and it resists only the PLA flexure's restoring force, not a spring.
- The spigot is printed at the SM1 tube nominal, which fits the SM1RC/M.

Hardware: 2 × FAS100, 2 × 98625A950 inserts, 2 × N52 10 × 5 × 2 mm magnet bearing pads (one under each FAS100 ball tip, §8.3), 1 × 2006N292 spring, 8 × M3 × 12 mm socket-head screws, 8 × M3 nuts, 8 × M3 flat washers.

Provisional: PLA creeps and relaxes under sustained strain, and the width stage is held deflected by its preload. Check the preload after the head has stood assembled for a few days, and again after warm storage.

8. Cutoff

The cutoff sits at the returned slit-image plane on the imaging rail. It is a rotating carriage (§8.1–8.5) holding interchangeable 64 mm cassettes (§8.6–8.7), which carry the cutoff element: a knife edge, a fine wire (§8.8), or a color filter (§8.9). The carriage model is `src/schlieren/parts/carriage.py`; the cassette blank and clamp bars are modeled in `src/schlieren/parts/cassette.py`.

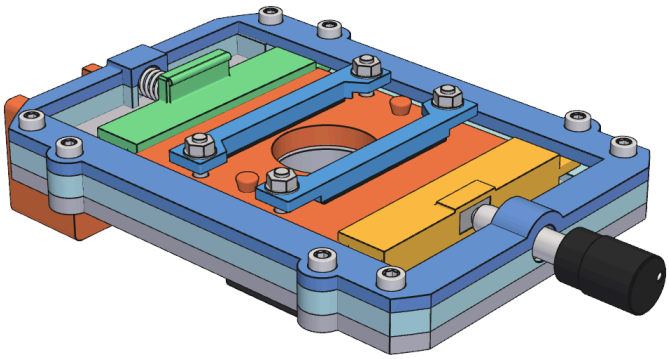


Fig. 8. Cutoff carriage with a seated cassette: guide frame, keeper frame and its retaining screws, driven and spring plungers, and the FAS100 fine adjuster

8.1 Spigot and rotation interface

The carriage mounts in 1 × Thorlabs SM1RC/M through a hollow printed spigot integral with the base plate, the same interface as the slit-head adapter (§7.6):

- Ø30.48 mm spigot (SM1 tube OD nominal), clamped directly by the SM1RC/M split ring (Ø30.7 mm bore, 10.16 mm thick, M4 locking screw across the split); the spigot end is flush with the ring's rear face;
- Ø24 mm light bore through the spigot, matching the plate aperture;
- Ø34 mm shoulder as the ring's axial stop: the ring seats against it, which places the plate back 12.0 mm in front of the ring face. Do not clamp the ring away from the shoulder.

The 12.0 mm ring gap is set by the rail shoe, not the post. At nominal zero the plate reaches about 63.5 mm below the optical axis, past the shoe top (about 52.4 mm below), so the plate back must clear the shoe end (15 mm from the post axis): calculated clearance 2.1 mm to the shoe and 10.7 mm to the post body, at every rotation. The shoulder clears the post top by 5.1 mm.

No SM1 tube, printed split clamp, retaining ring, or printed SM1 thread is used. The SM1RC/M provides the rotational bearing and lock: when loosened, the carriage rotates on its spigot. Nothing retains the spigot axially while the ring is loose, so support the carriage when rotating it.

Required carriage rotation:

- at least ±90°
- preferably ±100–120°

Define nominal zero as:

- cutoff line horizontal;
- fine-adjust translation vertical;
- fine adjuster pointing upward.

The cassette-loading side of the carriage faces the mirror, so cassettes are loaded and retracted from the forward side, away from the telephoto. The spigot, slip ring, and post are therefore aft of the base plate; the ring's rear face, 22.2 mm behind the plate back, is the rearmost part of the station on the optical axis and is what the telephoto front must clear.

Fore/aft optical positioning is done by moving the rail shoe.

Rotation envelope: the 2020 rail runs beneath the carriage at every rail position, 72.35 mm below the axis. The base plate, guide frame, and keeper are trimmed to a 70.35 mm radius about the optical axis (the rail top less 2 mm clearance). Only the outer plate corners are removed; the plate is otherwise 86 mm wide, widened to 94 mm at the side keeper-screw lugs (§8.4). Calculated from the CAD, the carriage with its FAS100 rotates clear of the rail over ±150°; only the FAS100 knob reaches the rail, near ±170–180°. Nothing projects below the plate at nominal zero (§8.5). **Provisional:** clearance to other equipment on the rail near the cutoff station is not yet checked.

8.2 Spigot print and fit

The base plate is printed deck down with the spigot up, so the spigot and shoulder need no supports; the guide frame above the deck is a separate print (the base/frame split exists for this). The three datum-pin shanks (§8.4) lie outside the shoulder and are flush-cut on the plate back.

The spigot is printed at the SM1 tube nominal, which fits the SM1RC/M without adjustment.

8.3 Fine adjustment

The carriage uses:

- 1 × Thorlabs FAS100, 1/4 in-80 × 1.00 in
- 1 × McMaster 98625A950 brass 1/4 in-80 insert
- 1 × McMaster 2006N292 compression spring

Insert geometry is taken exactly from the McMaster-Carr 98625A950 drawing, not from a generic 1/4 in-80 insert model. The drawing gives: internal thread 1/4 in-80; body outside diameter 0.313 in (7.950 mm); overall axial length 0.313 in (7.950 mm); under-flange/body length 0.298 in (7.569 mm); flange outside diameter 0.352 in (8.941 mm); flange thickness 0.010 in (0.254 mm); specified drill-bit size 0.313 in (7.950 mm); and minimum material thickness 0.298 in (7.569 mm). The 0.352 in dimension is the flange OD, not the bore or insert-body OD.

The insert bore is nominally 0.313 in (7.950 mm), matching the drawing. The printed bore is a pilot, finished after printing with a 5/16 in (0.3125 in) drill rather than relying on FDM accuracy; on the slit head this gave a good press fit (§7.7). The 9.3 mm insert support exceeds the drawing's 0.298 in (7.569 mm) minimum material thickness, leaving 1.7 mm of PLA beyond the end of the insert body (1.6 mm past the entry chamfer) and 3 mm of keeper crown over the bore. The spring plunger loads the FAS100 tip, so the screw pushes the insert outboard, toward the knob. The flange is therefore on the inner (cassette-side) face of the adjuster block, with the body running outboard: the reaction bears the flange against the block instead of relying on the press fit. The flange stands 0.254 mm proud of the flat block face; the plungers stop about 9 mm short of it. The insert is pressed in from the cassette side before the plungers are installed, with the flange bottoming on the face.

The adjuster axis is 8.77 mm above the plate back: the 8.94 mm flange clears the deck by 0.3 mm, so the flange seat is a plain face and the plate is not notched. The spring axis (§8.5) is the same line, so both plungers load the cassette along one axis.

FAS100 / insert travel envelope for CAD: the FAS100 uses a 1.000 in (25.4 mm) long 1/4 in-80 adjuster screw. The 80 TPI pitch gives 0.0125 in (0.3175 mm) axial motion per revolution. Using the insert's 0.313 in overall axial envelope as a conservative full-engagement allowance leaves approximately 0.687 in (17.45 mm) of theoretical screw motion. That is a geometric clearance estimate, not a published FAS100 travel specification. The carriage's working translation is ±5 mm / 10 mm total, leaving useful adjuster margin before either end stop.

The FAS100 ball tip does not bear directly on printed PLA. The driven plunger carries a small hard bearing insert made from a nickel-plated rectangular NdFeB magnet:

- Amazon B0DMCY4FN1
- N52 Bar Magnet - 10 mm L × 5 mm W × 2 mm H
- one magnet in the driven plunger

The magnet is used as a hard, wear-resistant plated bearing surface; its magnetic function is incidental. It sits in a close-fitting printed pocket with its rear face fully supported by PLA, with a small CA or epoxy tack for retention. The FAS100 ball contacts the broad plated face rather than bare PLA.

The plunger top is held near 12.0 mm by the cassette clamp-bar relief, which leaves only 0.6 mm of roof over the pocket (top at 11.37 mm). A local crown on the driven plunger over the pocket, 5 mm long from the outer face and 1.5 mm wider than the pocket on each side, restores a 1.3 mm roof (top at 12.67 mm). It is at the outboard end, clear of the clamp bars; it is blended into the plunger top by a 0.4 mm root fillet on its sides and inboard end.

The magnet pocket, with its crown roof and fillet, holds the magnet and the FAS100 ball bears on it smoothly. The insert's press fit and its flange seat on the flat block face are good.

8.4 Cassette seating

Cassette motion is controlled by opposed plungers:

- fine adjuster on one side;
- spring-loaded plunger opposite;
- three fixed rear datum contacts on the carriage define the cassette optical plane;
- shallow cassette/plunger bevels generate axial seating force against the datum contacts.

The three rear datum contacts are McMaster 97936A109 brass escutcheon pins / decorative finishing nails, used head-first as small domed sliding contacts. Each pin is approximately 3/8 in long, with a 0.050 in (1.27 mm) smooth shank and a measured 0.093 in (2.36 mm) diameter, 0.038 in (0.97 mm) high domed head; the head height sets the cassette's standoff from the datum face.

Mount each pin through a through-hole in the printed carriage with the domed brass head on the cassette side. The holes are modeled Ø1.75 mm: the measured 0.050 in (1.27 mm) shank, plus a 0.38 mm diametral allowance for the undersize of printed holes, plus 0.1 mm fit clearance; the holes take the shanks as a slip fit, so the CA is their only retention. If a shank binds, finish the hole with a #54 (0.055 in) drill. The hole fit only locates the shank; the head sets the datum. Seat the underside of the head firmly against the carriage datum face so datum height is established mechanically rather than by adhesive thickness. Wick a small amount of CA around the shank from the back for retention, then flush-cut and dress the projecting shank at the back of the carriage. Keep CA out from beneath the head/contact surface.

The three pin heads form a broad triangular support pattern around the optical aperture. The cassette rear face is the sliding datum surface and provides unobstructed swept tracks over the three domed contacts throughout the fine-adjust travel. The pin coordinates, set in the carriage CAD, keep every datum contact out of the swept track of the cassette's rear M3 clamp-screw countersinks over the full travel, so the datum tracks stay on uninterrupted printed rear-face material at every cassette position.

The plunger bodies are constrained to the intended translation axis by ears/tabs riding in slotted guiding rails. The guide slots prevent unwanted lateral movement or rotation while permitting the required axial plunger travel.

The ears run with only 0.15 mm of clearance below and 0.25 mm above, so their exposed edges are broken against binding: a 0.4 mm chamfer on the bottom edges, which take the printed first-layer squash, a 0.5 mm fillet on the top edges, and 0.6 mm fillets on the outer vertical corners (larger than the two edge breaks they meet). Where each ear leaves the plunger body wall, a 0.8 mm concave fillet on both the top and the underside relieves

the stress: the underside root is on the tension side under the seating reaction (about 10 N per ear, a few MPa nominal). It lies at x = 29.0–29.8 mm, clear of the track floor (x ≥ 32.3 mm) and the keeper opening edge (x = 33 mm).

The plunger bodies themselves run 0.3 mm above the base plate deck, so their underside perimeter carries a 0.4 mm chamfer, a lead-in over deck seams and a relief for first-layer squash. On the spring plunger it continues round the feet of the spring cup, including the edges of the pocket's opening toward the deck; the cassette-side bevels and lips, which generate the seating force, are left as designed.

The plungers need an assembly path into their guides: both longitudinal ends of the guide path are occupied, by the FAS100 adjuster support on one side and the spring end wall on the other. The guide tracks are therefore open from the cassette-loading side; the plungers drop into them, and a separate printed keeper frame closes only the guide-ear regions.

The keeper is a perimeter frame, not a solid lid: the central upper aperture must remain open for cassette insertion and removal. Its side members span the plunger guide tracks and retain the guide ears against upward escape while the printed track walls carry the lateral guidance loads. The frame may be locally scalloped or notched where the FAS100 adjuster axis intersects its envelope, provided the material directly over the swept guide-ear paths and the load paths to the fasteners remain intact. The keeper's spring-end member also carries the roof of the end-wall spring cup (§8.5).

Keeper retention is mechanical, with eight fasteners per carriage. Each is the existing common McMaster 91292A114 M3 × 0.5 × 12 mm socket-head screw with a McMaster 91828A211 full-height M3 × 0.5 hex nut (5.5 mm across flats × 2.4 mm high), captive in a hex pocket opening from the plate back. Screw heads and nuts bear directly on the PLA, without washers; tighten only to a light snug, since the joint needs little clamp and a firm hex-key torque can crush the bearing face.

- Four end screws, at (±38.5, −53) and (±38.5, +52) mm in the carriage frame (y along the travel, +y toward the FAS100). Each sits on solid fill closing its guide track from 1 mm beyond the outermost ear position to the plate end. The fill is also the plungers' end stop: 1 mm past full loading retraction for the spring plunger and past +5 mm travel for the driven plunger. These positions keep each nut pocket at least 1 mm inside the rotation envelope (§8.1).
- Four side screws, at (±42, ±37) mm, beside the middle of each plunger's ear travel. Each sits in a Ø10 mm round lug that widens the base, frame, and keeper locally to 94 mm, clipped at the guide-track wall.

The side screws carry the ear wedge reactions (below) into the frame close to where they arise. A first-order beam estimate, with each keeper side member simply supported between adjacent screws, gives at most about 0.07 mm of upward keeper deflection at the ears at the maximum-compression load (with a PLA modulus of about 3000 MPa assumed). That is within the 0.25 mm free play between the ears and the keeper; with only the four end screws it would be about 1.5–2 mm. The keeper side members are 3.0 mm thick; a printed keeper holds under the ear reactions without spreading. A keeper with 5.0 mm side members fits the same screws if more stiffness is wanted: its counterbores are deeper, so the head seat height is unchanged.

The screw heads sit in an approximately 0.5 mm-deep recess in the keeper, and the captive hex pockets are approximately 2.5 mm deep. This leaves approximately 1.5 mm of PLA above each nut pocket and gives nearly full use of the 12 mm screw length, with only slight nominal screw projection beyond the nut. The keeper seats on broad printed lands, so the fasteners clamp it against the carriage rather than suspending it between isolated points.

The vertical guide stack is: 4.0 mm carriage back plate; 2.0 mm rise from the plate front to the lower guide-ear running surface; 3.0 mm plunger guide ears in a 3.4 mm guide channel; and a 3.0 mm keeper frame.

The guide stack gives smooth plunger travel over the full range in both directions, and the ears slide freely in their channel; the captive nut pockets, keeper screw-head recesses, and datum pins all seat well.

Use a very light film of plastic-safe lubricant on the PLA-on-PLA sliding interfaces, particularly the plunger ears/tabs in their guide slots and any other plunger sliding faces. The lubricant is McMaster 1418K51 clear silicone grease, NLGI 1. Smooth/dress the printed sliding surfaces first, apply only a trace amount, cycle the mechanism, and wipe away visible excess so the lubricant acts as a thin boundary film rather than a grease-packed slide. Keep lubricant away from CA/epoxy/RTV bonding surfaces and apply it only after nearby adhesive work is complete.

The plunger seating lips and the cassette edge bevels are 27.5° from the cassette plane, with a 1.25 mm bevel depth.

For keeper-frame sizing, McMaster lists the 2006N292 spring rate as 0.14 lbf/mm. At the 9 mm fiducial compression this corresponds to approximately 5.6 N of in-plane plunger force; at the 14 mm most-compressed end of normal travel it is approximately 8.7 N. A frictionless first-order wedge model at 25–30° from the cassette plane gives approximately 9.7–12.0 N upward reaction per plunger at fiducial and approximately 15.1–18.7 N per plunger at maximum normal compression. These reactions are carried upward through the plunger ears into the keeper frame; about 20 N per plunger is the working-load scale before safety margin.

A printed cassette seats flat on the datum pins and is retained firmly by the plungers on the 27.5° bevels.

8.5 Spring seating and retraction

The spring plunger is preloaded by the McMaster 2006N292 compression spring (§8.3), acting between the frame's spring end wall and the plunger's outboard face on the adjuster axis, 8.77 mm above the plate back. No guide rod, retract handle, or penetration of the end wall is used; a guide rod through the end wall would project below the axis past the rail top at nominal zero.

Each spring end sits in a printed cup, 3 mm deep, sized to the measured 0.272 in (6.91 mm) coil OD plus 1.0 mm diametral clearance:

- one cup extends inboard from the end wall: its side walls are on the guide frame, and its roof is carried by the keeper's spring-end member, so the keeper closes over the spring end when installed. The cup's boss (11.5 mm wide, roof at 13.9 mm) rises 1.5 mm above the end member and continues back over its full depth, so the coil's end face bears on a wall as tall as the cup; 0.6 mm fillets blend its sides into the end member top;
- one cup extends outboard from the spring plunger;
- both cups are open toward the plate deck, which carries the coil. Their side walls and roofs locate the coil ends laterally and against escape toward the cassette-loading side. Both print with legs on the bed and the roof bridging the coil, without supports.

The cup roof is 1.2 mm over the coil and the side walls 1.8 mm, so the feet keep a 1.0 mm flat inside the 0.4 mm underside chamfer (§8.4).

The spring's free length is about 4–4.6 times its mean coil diameter, and both ends sit on parallel faces because the plunger is guided. That is below the roughly 5.3 critical ratio for buckling with fixed ends, so the cups need only locate the ends rather than guide the full length. The spring itself does not guide the plunger; the slotted carriage guides constrain plunger translation and rotation.

The spring has a free length of approximately 25.5 mm. The carriage is designed around a 16.5 mm spring length at the fiducial / mid-travel cassette position, corresponding to approximately 9 mm spring compression from free length.

Normal supported plunger travel is ±5 mm about the fiducial position, for 10 mm total working travel. The corresponding spring-length range is approximately:

- 21.5 mm at the least-compressed end of normal travel;
- 16.5 mm at fiducial / mid travel;

- 11.5 mm at the most-compressed end of normal travel.

Loading retraction is a further 6 mm from fiducial (10.5 mm spring length), leaving 4.5 mm between the cups. Any additional compression capacity is treated as design margin rather than normal commanded travel. The spring plunger's ears reach the end of their track before the spring is compressed solid, so the track end stop, not the spring, limits retraction.

Retraction for cassette loading uses a thumb tab on the spring plunger's front face at its outboard end:

- 20 mm wide × 3 mm thick × 6 mm tall, centered between the guide ears and outside the cassette footprint;
- the thumb reaches over the cassette and pushes the tab's inner face outboard, at about 9 N at full retraction;
- 0.5 mm chamfers on the exposed edges and 0.5 mm root fillets on its thumb-side face and ends; the spring-facing face is left square;
- a Ø1.6 mm grip bead along the inner top edge, flush with the tab top, prevents finger slip.

The tab is kept low, close to the plane of the guide ears, so that pushing on it does not twist the plunger and jam the ears.

8.6 Common cassette standard

Cassette envelope:

- 64 × 64 × 5 mm PLA
- 24 mm diameter central clear aperture
- identical shallow bevels on all four perimeter edges
- insertable at 0 / 90 / 180 / 270°
- perimeter kept clear enough to use any opposing edge pair
- orientation/polarity indicated by markings rather than a mechanical key.

Cassettes load with the spring plunger retracted 6 mm beyond fiducial (§8.5).

Cassette variants, all built on the common blank:

1. single razor knife edge;
2. centered fine-wire or filament dark-field cutoff (§8.8);
3. striated / multicolor gel filter (§8.9);
4. dark-center color filter (§8.9).

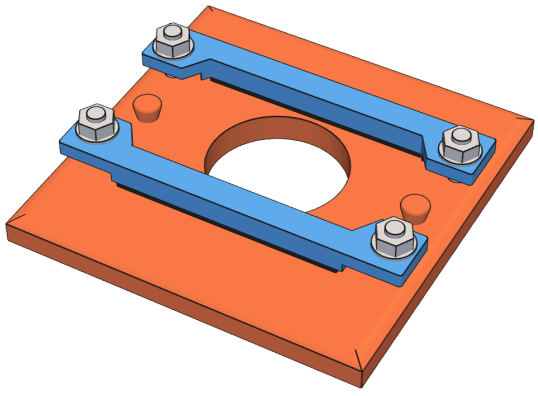


Fig. 9. Cassette blank with its two clamp bars and hardware

The standard cassette blank carries one opposed pair of integral filament cleats on the front / non-datum face, rather than making cleats a special cassette variant: two cleats, one on each side of the 24 mm optical aperture, at (±25, 0) mm on an aperture centerline, outboard of the razor blades and clamp bars. The cleats are slightly conical, Ø3.5 mm at the cassette face widening to Ø5 mm toward the top and rising 3 mm, so filament tension drives fine wire or thread down against the cassette front face. The tops are rounded to avoid damaging fine wire. Route a filament in an S path, around opposite sides of the two cleats; the free span then crosses approximately through the aperture centerline. Small angular error from finite cleat diameter or wrapping is acceptable because the SM1RC/M rotates the complete tube/cassette assembly. This makes the same universal cassette blank usable for blade, filter, Photofoil, wire, and thread experiments while keeping the rear datum face unobstructed.

Provisional: cleat geometry is pending a wire-mounting trial.

8.7 Cassette clamp bars

These bars and their hardware are the clamping standard for the cassettes (§8.6); the slit head uses its own smaller bars (§7.5).

Common clamp bars:

- two identical printed rigid PLA bars fit each cassette;
- each bar uses two fasteners, one toward each end, so clamp force is distributed along the element;
- no holes are made through razor blades, filter material, or other optical elements;
- no blade-specific locating pocket or guide is required;
- McMaster 9852N37 adhesive-backed 1/32 in, 60A solid EPDM is applied to the removable clamp-bar face to provide compliance and friction;
- the bar bears primarily on the broad flat portion of a razor blade rather than relying on the folded spine for its clamp datum;
- the same clamp-bar geometry holds razor knife edges, Roscolux gel/filter material, Photofoil, and similar thin sheet cutoff elements.

Clamp-bar fastening is from the cassette rear/datum face toward the optical/front face so no screw ends or nuts can protrude from the sliding/datum surface. Fasteners are positioned outside the optical-element footprint where practical; the clamp bar spans and pinches the element against the cassette front face. At each fastener, use:

flush M3 countersunk flat-head screw → 5 mm cassette body → printed clamp bar → M3 flat washer → ordinary M3 hex nut

The M3 flat-head screw is seated fully flush or very slightly below the rear cassette face and retained in its countersink with a small CA tack using McMaster 1818A45 / Loctite 4061. The CA serves only to keep the screw captive and resist rotation while the front-side nut is adjusted; it is not relied upon as the structural clamp load path. The ordinary M3 nut remains exposed and accessible on the front of the clamp bar. Place the washer between the nut and printed bar. Do not use prevailing-torque/nyloc nuts or threadlocker here.

Cassette clamp-bar hardware

Component	McMaster part	Specification
Flat-head screw	92125A133	M3 × 0.5 × 14 mm, fully threaded 18-8 stainless, 90° countersunk head; 6 mm head diameter, 1.7 mm head height, 2 mm hex drive
Flat washer	93475A210	18-8 stainless, 3.2 mm ID × 7 mm OD × 0.4–0.6 mm thick; DIN 125 / ISO 7089
Ordinary full-height hex nut	91828A211	M3 × 0.5, 18-8 stainless, 5.5 mm across flats × 2.4 mm high; DIN 934 (the common split-clamp nut, §5.3)

The clamp bars are 4 mm printed PLA. The cassette blank has four Ø3.4 mm screw holes at (±27, ±12) mm with rear-entry 90° countersinks matched to the 92125A133 head (Ø6.2 mm at the face: 6 mm head plus 0.2 mm diametral allowance), and each bar end provides a washer bearing land at least 7.2 mm across. The screw head finishes flush with or slightly below the rear/datum face so it cannot contact the carriage datum pins or interfere with cassette sliding. The fasteners, bars, nuts, cleats, and projecting screw ends are clear of the 24 mm optical aperture and carriage interfaces. The holes and countersinks are features of the universal cassette blank, not blade-specific geometry.

Because the clamp fasteners do not pass through the optical element, the screw length is not sized from a blade-thickness stack. The 14 mm length gives full nut engagement and holds razor blades under the 4 mm bars, with the head seated flush in its countersink. The screws are vertical and the lips act at the cassette edges, so the screw length does not bear on the lips; the bar ends, which reach the bevel zone, are relieved to clear them.

8.8 Fine wire

The dark-field filament set is enameled copper magnet wire in three readily available AWG sizes. The nominal diameters below are bare-conductor diameters; enamel makes the finished optical obstruction slightly larger.

Wire size	Nominal bare diameter	Approx. metric diameter	Experimental role
44 AWG magnet wire	0.0020 in	0.0508 mm	fine / high-sensitivity filament
34 AWG magnet wire	0.0063 in	0.160 mm	intermediate filament, near the expected slit-image width
30 AWG magnet wire	0.0100 in	0.255 mm	coarse filament for stronger gradients / deeper dark field

Exact finished outside diameter is not critical for the qualitative instrument.

The universal cassette cleat pair described in §8.6 is the quick-change mounting for these wires and for dark sewing thread or other fibers. Apply only modest tension, particularly to 44 AWG wire; the conical cleats seat the filament against the cassette face, and the S-wrap places the free span approximately on the aperture centerline.

8.9 Color-filter material

Filter material is cut from the Roscolux 3 × 6 in Designer Color Selector, which provides more useful material per color than very small swatches.

For color schlieren:

- use the white Nichia LED;
- lock camera white balance;
- lock exposure.

9. Camera and telephoto support

The camera and lens are chosen (§9.1–9.2). Their support is a lens cradle and a phone rest (§9.4–9.8), with an alignment procedure (§9.9). **Provisional:** the phone rest's retainer and safety catch, the fit allowances of the printed yokes and of the rod groove, which are still to be tested, the inside of the phone case, and the alignment procedure (§11).

9.1 Camera

- Apple iPhone 17, base model, in intended protective case
- held in landscape, with the rear cameras toward the mirror and the camera end of the phone toward the optical axis
- the Main camera is used, at 1× (or its 2× crop). It is the lower of the two rear apertures with the phone in portrait, "rear camera 2" on Apple's drawing (`docs/reference/Apple-iPhone-17.pdf`); the upper aperture is the Ultra Wide, used only at 0.5×
- measured camera optical axis approximately 60.6 mm above the lower long edge, in the case
- measured Main camera axis approximately 1.350 in / 34.3 mm from the near short edge, in the case; the case shape makes this hard to measure accurately with calipers
- measured case thickness approximately 0.510 in / 12.95 mm (bare phone 7.95 mm); the case corners are rounded
- measured mass 188.7 g bare, with its glass screen protector, and 238.6 g in the case

Manufacturer dimensions of the bare phone, from the drawing:

- both rear cameras are 13.62 mm from the nearer long edge, on a line parallel to it, so the measured axis height applies to either aperture
- the Main camera is 31.34 mm from the top short edge, 17.72 mm beyond the Ultra Wide at 13.62 mm
- body 149.61 × 71.45 × 7.95 mm; the camera glass stands 3.45 mm above the back glass

The drawing and the two measurements agree. The cameras are 57.83 mm from the farther long edge of the bare phone, so the measured 60.6 mm implies a case wall of about 2.8 mm at that edge; the 31.34 mm and the measured 34.3 mm imply about 3.0 mm at the short edge (both calculated).

Phone model, from Apple's drawing and the measurements above: the bare body with the drawing's corner profile and glass edge, the camera plateau (1.78 mm above the back glass; its top face is blended into the glass by a concave fillet 2.62 mm wide, whose foot is 1.18 mm and 1.34 mm from the long and short edges) with the two Ø16.00 camera glass rings (3.45 mm; the rings are 17.72 mm apart and do not touch), and the side, action, and volume buttons (0.45 mm proud). The Camera Control and SIM tray are flush with the edge and have no geometry. The case is an OtterBox Commuter, whose outside is the measured size: 155.5 × 77.0 mm, with walls of 2.96 mm at the short edges and 2.77 mm at the long edges calculated from the measured camera position, and 12.95 mm thick. **Provisional:** OtterBox publishes no drawing, so the inside of the case is assumed apart from the front rim, which is measured to stand 0.050 in. (1.27 mm) proud of the screen. That puts the bare phone's camera glass 0.28 mm under the camera lip, whose top is the back plane where the lens seats, and leaves the rest of the 12.95 mm to the back plate's inside; the opening's corner radius (12 mm) and the 1.2 mm round on the front rim's inner edge, which the rim height limits, are assumed. The camera opening is measured: 1.585 × 1.35 in. (40.26 × 34.29 mm), centered on the plateau along the length, with its long edge 0.180 in. (4.57 mm) from the case's camera-side edge; it clears the plateau and the flash. The model does not show the case's port openings.

Initial recording modes:

- 4K/30 for normal work;

- 4K/60 for faster flow phenomena;
- SDR / HDR off as a stable starting point.

9.2 Telephoto

- iOgrapher proSnap 7×
- 17 mm camera-standard mount
- approximately 37.0 mm barrel diameter
- measured mass approximately 215 g

Measured sections, from the aft end forward:

Section	Length	Outside diameter
threaded boss ring (17 mm)	0.10 in / 2.5 mm	17 mm thread
shoulder	0.40 in / 10.2 mm	1.26 in / 32.0 mm
main barrel	4.285 in / 108.8 mm	about 37.0 mm

The overall length is 4.785 in / 121.5 mm. The boss threads into the lens mount on the phone case, and the shoulder seats against the mount; the model takes the aft end of the boss to be at the back plane of the cased phone. The main barrel is 1.457 in / 37.01 mm in diameter at its rear end and 1.459 in / 37.06 mm further forward. The focus ring, 0.455 in / 11.6 mm long, ends 0.175 in / 4.4 mm behind the front of the barrel, so it starts 105.5 mm from the aft end. The barrel is clampable from 12.7 mm to the focus ring, about 93 mm, and the focus ring must stay clear. The overall length is about 4.7 mm more than the 116.8 mm listed for the lens; the difference is not resolved.

The lens is modeled in `src/schlieren/parts/lens.py`, for visualization only.

The lens barrel, rather than the phone shell, is the preferred transverse optical datum because accurate lens centering minimizes vignetting.

9.3 Alignment requirements

The returned slit image at the cutoff is the aperture stop for the camera: every ray from the test region passes through it, then spreads again at the ±1.82° cone angle (§3.2). The values below are calculated, and are **provisional** where they rest on assumed values: a lens front about 42 mm behind the cutoff plane, a 10 mm lit slit image, a Main-camera pupil of about 3.7 mm (6 mm focal length at f/1.6), and a telephoto clear aperture near 30 mm.

Quantity	Value
Beam cone at the lens front	about 2.7 mm diameter
Beam footprint at the lens front	about 2.7 × 12.7 mm
Slit image at the phone, reduced 7× by the lens	about 1.4 mm long
Frame at 1×, 4K 16:9, through the lens	about 9.9° × 6.1°
Mirror in the frame	3.6° of the 6.1° height

The resulting tolerances on the camera and lens as one body:

- centering on the beam: about ±8 mm along the slit direction and ±13 mm across it, before the reduced slit image leaves the phone pupil; the lens front aperture allows a similar margin;

- pitch: about ±1.2°, the spare frame height either side of the mirror;
- yaw: about ±3°.

Pointing is therefore the governing adjustment, and centering is forgiving. Image quality still favors centering well inside the vignetting limit.

9.4 Support

The phone and the lens threaded onto its case are one rigid body. The lens barrel is its datum (§9.2): the lens is held at two stations along its barrel, and the phone hangs from its lens mount. Two separate rail fixtures carry them:

- the lens cradle, specific to the lens, holds the barrel at both stations on four adjusting screws (§9.5) and is held down by two retaining bands (§9.8);
- the phone rest, specific to the phone, supports the phone's lower edge at one point (§9.6). A different phone needs only this part redesigned.

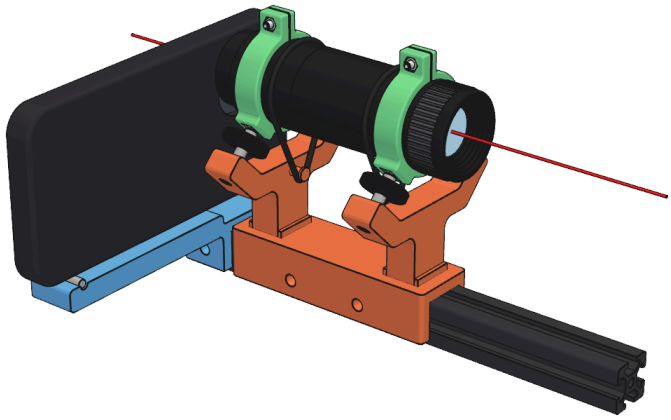


Fig. 10. Lens cradle and phone rest on the imaging rail, with the phone and the lens

The body is located by six contacts and no more, so adjusting one station does not strain the lens mount:

- four screw tips, two at each station, fix the lens axis in x and z at both stations, and so its position, pitch, and yaw;
- one of those tips sits in a groove, fixing the lens fore/aft;
- the phone rest fixes roll.

Each station's pair of screws places one point of the lens axis, so the two stations give position and pointing without coupling between them beyond the lever ratio. A second groove would fix the fore/aft position a second time against the spacing of the collars on the lens, and would stop the pad sliding that yaw adjustment needs. Holding the lens, not the phone, takes the lens weight off the lens mount: the mount carries about 1.3 N in shear (calculated), against roughly 170 N·mm of bending if the lens were cantilevered from the phone.

The model is `src/schlieren/parts/camera_support.py`. It holds the printed parts, with their fits, pockets, fillets, and wall thicknesses, and checks contacts, clearances, and the lens and cord loads; purchased hardware is envelopes or vendor models. The phone is modeled in `src/schlieren/parts/phone.py` (§9.1).

9.5 Lens cradle

The cradle is five printed parts and two bands. Two split collars clamp the barrel; two yokes carry the adjusting screws; one shoe straddles the rail and carries both yokes.

Collars

Each collar is a thin ring on the barrel with a clamp ear on top and a flat pad platform at each screw:

- the aft collar centered about 24 mm from the lens aft end, which leaves finger room for its thumb nuts ahead of the phone; the fore collar centered about 96.5 mm from the aft end, with its front face about 3 mm behind the focus ring; about 72.5 mm between them;
- 12 mm wide, with a 3 mm wall round a bore of 37.3 mm, the measured 37.0 mm barrel plus a 0.3 mm diametral allowance, so a collar is 43.3 mm across. The wall is thin because the pad platforms carry the local thickness; the pad loads are a few newtons, so stiffness is not limiting;
- the ear takes the common M3 split-clamp hardware (§5.3) with the rail shoe's split and ear thicknesses: the screw passes through the thin ear on the left, and the hex nut is captured in a pocket in the thicker ear on the right, open to the outside, with a roof of at least 1.5 mm over it. The screw axis stands 4.25 mm above the ring top, so the screw head clears the ear's root fillet;
- inside corners, the ear and pad platform roots, are filleted at 2 mm for stress control; the other outside corners and the outer edges of both ring ends at 1 mm, for the fingers. The faces of the split stay sharp so that the gap stays fixed.

The two collars are the same part except that the aft collar's left pad is a rod seat in place of a magnet seat (below).

Pads and pockets

Each screw tip bears on a flat pad on the collar, tangent to the ring at the screw: the pad face stands one magnet thickness outside the ring, and the pocket floor is tangent to the ring's outside. Each pad sits in a pocket with a rim round it, so a ball end that wanders reaches a wall instead of the edge of the pad:

- three pads are the 10 × 5 × 2 mm magnets, set with the 10 mm side across the rail, bonded into pockets 0.2 mm larger across each dimension (0.1 mm per side, the allowance every printed magnet pocket in the design uses);
- the fourth, at the aft left screw, is a groove between two dowel pins of 3 mm diameter and 10 mm length, lying across the rail on the pocket floor. The ball sits between them, free to slide across the rail but not along it. A printed rib 1.0 mm high and about 1.7 mm wide runs between the pins along the pocket, and each pin lies against its top corner: the rib sets the pins' spacing (4.5 mm between axes) and keeps them parallel, so bonding them needs no spacer. The ball stays 1.0 mm clear of the rib;
- the rim stands 0.8 mm above the pad face. The ball-tip screw's end face is 1.2 mm behind the ball apex, so the rim stays below it;
- above the pad face the pocket opens outward by 0.23 mm per side, so that the rim edge stops the ball apex 1 mm short of the pad edge, the margin the travel limits below assume. The walls round the pockets are 1.5 mm.

The two screws of a station are at right angles, so advancing one slides the collar across the other's pad by the same distance. Pad size, not screw travel, limits the adjustment. Calculated, keeping each ball 1 mm inside the pad edges, which the rim enforces:

Adjustment	Range
Each screw	about ±3 mm
A station, in x or in z alone	about ±4 mm
Lens tilt, pitch or yaw	about ±2°
One turn of both screws of a station	about 1.1 mm, or about 0.9°

The tilt limit comes from the 5 mm side of the pads, on the pads that slide along the rail as the lens yaws about the groove. These ranges cover the alignment tolerances (§9.3) and the expected build errors.

Yokes

Each yoke is one printed part, 14 mm thick along the rail, in the shape of a T with an inclined arm either side, printed on its side so that every layer lies in the plane of the loads:

- a column 20 mm wide, the width of the rail, rising from the shoe to a crossbar 10 mm thick whose top stands 13 mm below the bottom of the collar, clear of the thumb nuts;
- each arm is a slab 10.5 mm thick, square to its screw. Its lens-side face carries the insert and runs down to the crossbar top; its back face runs down to the crossbar underside, so there is no stub or step;
- fillets of 5 mm at the column root, 4 mm where the arm blends into the crossbar, and 2 mm round the outer arm corners;
- each arm holds a heat-set insert (§9.7) in a 6.4 mm hole from the lens-side face with the flange recessed flush. The wall round each insert and its flange recess is at least 2.9 mm, and a 5.5 mm bore carries on through the back of the arm for the screw and its hex key.

Under the adjuster reactions of about 8 N the crossbar root sees about 0.6 MPa and the arm tip moves about 0.01 mm (calculated, with a PLA modulus of about 3000 MPa assumed), so stiffness and strength are not limiting; the fillets are margin.

Shoe and joint

The shoe's lower section is that of the common rail shoe (§5.3): a 20.4 mm rail opening, 5 mm side walls, 18 mm skirts, a 5 mm deck, and 2 mm rounded outside corners. M5 clearance holes cross the skirts at rail mid-height for the side-slot screws, two of them 36 mm apart midway between the collars. The shoe extends 9 mm beyond each collar face, so it is symmetric about the middle of the two stations. Each yoke is bolted to the deck, so that it prints on its side and the shoe prints without supports:

- one 90° countersunk M5 screw from the underside of the deck, on the rail centerline, with its head flush with the underside, into a heat-set insert in the column foot, with the insert's flange bearing on the deck;
- two straight walls on the deck across the rail, ahead of and behind the column, 1.6 mm thick and 3 mm high, with 0.2 mm clearance to the column faces. They locate the yoke along the rail and stop it twisting;
- the screw ends inside the insert with about 5 mm of engagement.

Provisional: the groove between the rods and the ball-stop rims, and the insert holes and locating walls of the yokes, are untested; the magnet and rod adhesive is not chosen; the fitting and removal sequence is set by the bands (§9.8).

9.6 Phone rest

The phone hangs from the lens with its body to the outboard side of the optical axis (§3.4). Its weight, centered about 43 mm from the lens axis, would turn the lens in its collars. The phone rest stops that: the phone's lower long edge sits on one dowel pin lying along the rail, about 100 mm outboard of the axis, on the arm of a short printed shoe of its own. The shoe's lower section is that of the cradle shoe (§9.5), with a single pair of M5 clearance holes for the side-slot screw.

- The shoe is 22 mm long and centered under the rod. The arm runs across the rail from the shoe's outboard wall to 8 mm beyond the rod, 10 mm deep and as long as the shoe. The rod lies half sunk in a half-round seat in its top, with 0.1 mm radial clearance, and is bonded.
- The part is printed on its side, like the yokes: it is one profile in the plane of the load, extruded along the rail, so every layer is that profile and the arm needs no bridge or support. Only the side-slot screw bores are horizontal. The shoe's vertical corners on the arm side are left sharp, where the arm joins the wall.

- The arm's inside corners are filleted against stress: 3 mm under the arm where it meets the shoe wall, the largest because it takes the bending, and 2 mm where the arm top steps up from the deck. The outside corners at the tip and the step are rounded 2 mm, and the seat edges 0.5 mm. The arm is kept 10 mm deep so that the root fillet stays at least 1 mm above the side-slot screw head on the outside of the skirt.
- With the measured 60.6 mm camera-axis height, the phone's lower edge is about 11.75 mm above the rail top.
- The rest carries about 1.0 N (calculated, same assumptions as the contact loads in §9.8). It matters to the cradle, because it keeps the fore pad load from falling toward zero.
- The rest is fixed. Raising or lowering the aft collar rolls the phone slightly about the lens axis, which does not matter; the phone is not gripped and slides on the rod.
- The rest holds no optical alignment, so its dimensions are not critical.
- Because the rod lies along the rail, the rest needs to be only roughly under the phone.

Provisional: a light retainer to keep the phone on the rod, and a loose safety catch against a knock, are not yet designed.

9.7 Adjuster hardware

There are four adjusting screws, two under each collar. Each is a ball-tip set screw in a heat-set insert in its yoke arm, with a thumb nut threadlocked mid-screw as a handwheel.

Hardware:

- McMaster 93339A252, M5 × 0.8 × 25 mm ball-tip set screws: alloy steel, 25.0 mm overall including the ball, 2.5 mm hex socket, and a 3.0 mm ball (ball size from the vendor model, `cad/vendor/McMaster-93339A252.step`; drawing `docs/reference/McMaster-93339A252.gif`)
- McMaster 94459A797, M5 flanged heat-set inserts: 6.73 mm long overall, with a 7.92 mm flange 1.02 mm thick, a 7.11 mm knurled body, and a 6.32 mm pilot (drawing `docs/reference/McMaster-94459A797.gif`; vendor model `cad/vendor/McMaster-94459A797.step`). The drawing calls for a 6.40 mm hole and at least 6.48 mm of material. Each is set from the lens-side face of its arm, flange outward and recessed, so the screw's reaction presses it into its hole. Two more are set in the yoke feet for the joint screws
- McMaster 92125A208, M5 × 0.8 × 10 mm 18-8 stainless hex-drive flat head screws (90°, 10 mm head), two for the yoke joints (§9.5)
- McMaster 92815A202, M5 thumb nuts used as threadlocked mid-screw handwheels
- Loctite 271, McMaster 91458A160
- heat-set installation tip 92160A327;
- Amazon B0DMCY4FN1, N52 Bar Magnet - 10 mm L × 5 mm W × 2 mm H, as hard plated bearing pads, one at each of three screws;
- McMaster 91585A351, Ø3 × 10 mm 18-8 stainless dowel pins (ISO 2338 m6, chamfered ends), two for the groove
- at the fourth screw;
- McMaster 91585A457, Ø4 × 20 mm 18-8 stainless dowel pin, for the phone rest rod;
- the common M3 split-clamp screw and nut (§5.3), one pair per collar.

9.8 Retaining bands

The collars sit on their pads under gravity alone, so each pad pair is held down by an elastic band. The pairs carry only the lens and phone weight, and the fore pair least of all: contact loads, calculated from the measured 215 g lens and 238.6 g cased phone, with an assumed balance point 60–70 mm from the aft end, are

- aft screw pair: about 2.7–3.0 N;
- fore screw pair: about 0.5–0.8 N. The phone hangs behind the aft collar, so its weight pivots the lens about the aft screws and unloads the fore pair, but it cannot lift it: the lens moment is the larger.

A downward push on the phone, such as a screen tap, lifts the fore end when it exceeds roughly 2.5 times the fore pair's load, about 1.2–1.9 N unaided, and a sideways nudge unseats a pad when it exceeds the load on its pair. The alloy-steel screw tips are attracted to the magnet pads, which adds some preload; it is not relied upon until measured.

Each band is an endless O-ring, 2 mm in section, of oil-resistant 70A Buna-N with an inside diameter of about 40 mm, lying round the bare barrel beside its yoke on the side toward the other collar, 11.8 mm from the collar's center. Its two legs run down inboard of the thumb nuts to a peg on the center of the yoke's crossbar face. The peg has a 4 mm stem and a 6 mm lip; the ring lies on the stem under the lip.

- The band acts on the barrel, which the collar's clamp carries to the pads, exactly as it carries the lens weight. The ring needs no groove on the collar and settles into the plane through the peg.
- The legs are about 65° above horizontal, so 5 N of hold-down takes about 2.75 N in each leg. A ring of that size stretched about 19% pulls about 3 N (calculated, with a 70A rubber modulus of about 5 MPa assumed).
- Each band acts about 12 mm from its collar. With one band beside each yoke, between the collars, each pad pair receives the hold-down of 5 N in total from the two bands (calculated), about 4.2 N from its own band and about 0.8 N from the other. That raises the tap threshold at the fore end to roughly 13–14 N.
- The legs clear the thumb nuts by more than 5 mm, so the cord stays out of the way of the fingers.
- To fit, slide the rings over the front of the lens onto the bare barrel before clamping the collars on; refitting one afterwards means stretching it round the other collar and its ear. A band stays on the lens when the lens is lifted out with the peg end unhooked.

Provisional: the contact loads rest on the assumed balance point above, and the ring size is chosen on an assumed modulus; the next larger inside diameter is the alternative if the 40 mm ring is too tight.

9.9 Lens alignment

This is the alignment the cradle must make possible, kept here as a feasibility check on the adjusters; a consolidated setup and alignment procedure comes later. **Provisional:** the procedure is untested, and its targets assume that the printed shoe seats fully on the rail top and the collars seat on their barrel sections.

The tolerances (§9.3) are loose compared with the adjuster resolution, so the alignment is mechanical first and then finished through the camera, with no laser.

Each adjuster moves the lens axis along its own screw, so at one station, with the screws at ±45° and a thread pitch of 0.8 mm (calculated):

- both screws turned the same way, n turns: z moves n × 1.13 mm;
- the two turned in opposite directions, n turns each: x moves n × 1.13 mm;
- one screw alone, one turn: x and z each move about 0.57 mm;
- pitch and yaw come from setting the two stations 72.5 mm apart differently; 1 mm of difference is about 0.8°.

The sequence:

1. Fit both collars and the phone on the lens at their stations, set the four thumb nuts to equal exposed screw length (mid-travel), and note their positions. Put the cradle shoe on the rail with its clamp screws loose.
2. Slide the shoe along the rail to put the lens front at the planned gap from the slip ring's rear face, then snug the clamp screws. The groove at the aft left screw fixes the lens fore/aft on the cradle, so nothing later moves it.
3. Center the barrel at each collar mechanically, aft first, and re-measure the other collar after each change. Height: from the shoe deck, beside the yoke, up to the barrel top, whose target is 72.35 + 18.5 − 5 = about 85.85 mm (the deck top is 5 mm above the rail top; the barrel diameters of 37.01 and 37.06 mm differ by too little to matter). Lateral: with the shoe pressed to one side of the rail, taking up its 0.4 mm lateral play, from the shoe's outer wall to the barrel side, whose target is 18.5 − 15.2 = about 3.3 mm. A ±0.5 mm error between the stations is about 0.4° of pitch or yaw, within tolerance.
4. Finish through the camera, because the phone camera may not lie on the barrel axis in its mount and case. With the source lit and the cutoff cleared, view the returned field at 1× and trim the aft screws in x and z until the vignetting at the corners is even; a turn of one screw is only about 0.6 mm, so turn in small steps. The mirror should sit centered in the frame, which sets pitch and yaw to within the spare 1.2°.
5. Check the roll: the phone rest rod should just touch the phone's lower edge without lifting it, a paper strip sliding between them with light drag. A lower-edge height different from the model is taken up by shimming the rod seat.
6. Record the final thumb nut positions, so that refitting the lens needs only a check.

- Each station's range is only about ±4 mm, so a larger offset is taken up by moving the shoe on the rail, not with the screws.
- The aft thumb nuts must stay clear of the phone while they are turned; the model leaves finger room, to be checked on the real phone.

10. Mirror cell

10.1 Plates and basic architecture

The mirror-cell is constructed from three Baltic birch plywood plates:

- base plate: 11.5 in wide × 7.0 in front-to-back × 1/2 in;
- cell-adjuster plate: 11.5 in wide × 11.0 in high × 1/2 in;
- moving mirror plate: 3/4 in cut to the truncated-hex geometry below.

The base plate provides the structural/tripod-support base and is joined to the cell-adjuster plate by structural brackets described in §10.7. The cell-adjuster plate carries three fine-adjustment stations (§10.2) which couple it to the moving mirror plate. The 3/4 in moving mirror plate carries the 203 mm mirror and moves relative to the fixed cell-adjuster plate.

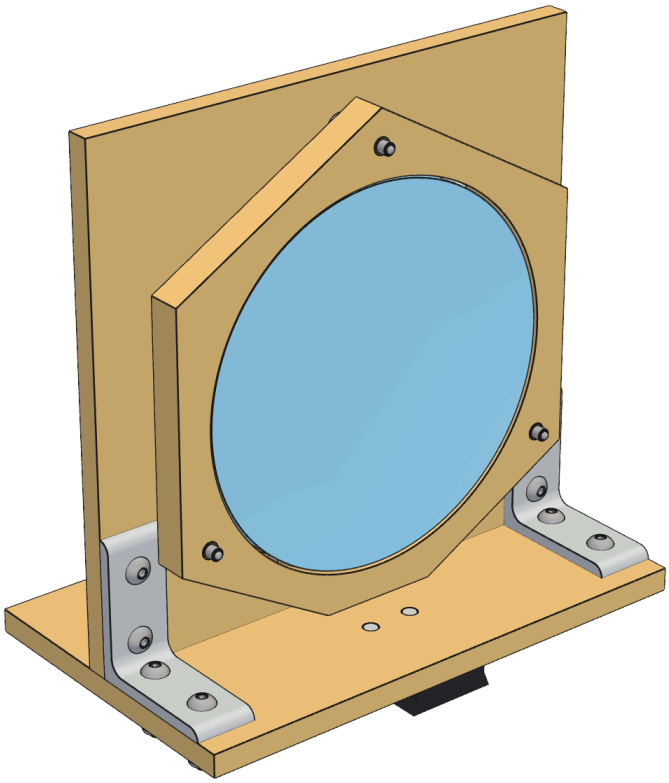


Fig. 11. Mirror cell from the mirror side: base plate, cell-adjuster plate, moving mirror plate and mirror, and corner brackets

The cell-adjuster plate is centered left/right on the 11.5 in-wide base. The rear face of the 1/2 in fixed cell-adjuster plate is 4.000 in aft of the front edge of the 7.0 in-deep base, leaving 3.0 in of base behind it. Two McMaster 8681N11 structural angle brackets face forward, toward the moving mirror plate; their measured forward footprint is approximately 3.5 in from the fixed-plate rear face, leaving approximately 0.5 in of base margin at the front edge. This orientation places the brackets on the mirror side of the fixed plate so they directly resist the forward overturning moment of the mirror assembly.

Using measured masses of 1.56 kg for the approximately 18 mm-thick mirror, 840 g for the assembled fixed + moving plates and adjuster hardware, about 318 g for the 11.5 × 7.0 × 1/2 in base plate, and 79 g each for the two aluminum brackets, the estimated completed mirror-cell fore/aft center of gravity is approximately 3.0 in / 76 mm

aft of the front edge of the base. A working uncertainty of approximately ±0.2 in / ±5 mm is conservative. This places the expected center of gravity approximately 1.0 in / 25 mm forward of the rear face of the fixed cell-adjuster plate. The 130 g CAMVATE removable quick-release plate is centered 0.5 in / 12.7 mm aft of this location, a small offset that has little effect; final balance is trimmed using the quick-release system's generous fore/aft sliding travel.

The cell uses:

- three fine-adjustment stations arranged at 120° intervals;
- compression-spring preload at each station;
- spherical washer interfaces for angular accommodation and gravity-load transfer;
- tripod interface through the CAMVATE 577/501/504-style quick-release system and the Sunwayfoto LB-68R leveling base.

The moving mirror plate geometry is:

- start from a regular hexagon 11.000 in / 279.4 mm vertex-to-vertex, oriented with one vertex upward;
- the corresponding normal flat radius / apothem is 4.7631 in / 120.98 mm;
- three alternating vertices lie on the three 120° adjuster axes, with the top vertex occupied by one adjuster axis and the two lower-side occupied vertices by the other two;
- the three adjuster-hole centers lie on a 4.750 in / 120.65 mm radius from the mirror center;
- the mirror aperture is 8-1/8 in / 206.38 mm diameter, providing approximately 1/16 in radial clearance around the nominal 203 mm mirror;
- clip the unused bottom hex vertex with a horizontal chord whose midpoint is 4.7631 in / 120.98 mm below the mirror center, matching the original hex apothem. This removes approximately 0.7369 in / 18.72 mm of the original lower tip and produces a new lower edge approximately 2.553 in / 64.8 mm long;
- the resulting trimmed moving-plate height is approximately 10.263 in / 260.7 mm from the new lower edge to the top vertex.

The nominal clearance from the bottom trimmed edge of the moving mirror plate to the top surface of the base plate is 3/8 in / 9.53 mm. This places the mirror center approximately 5.138 in / 130.51 mm above the base top. The top moving-plate vertex then lies approximately 10.638 in / 270.21 mm above the base top, leaving approximately 0.362 in / 9.2 mm of fixed-plate height above it within the 11.0 in-high cell-adjuster plate.

This truncated-hex geometry preserves the same approximately 0.701 in / 17.8 mm radial plywood width between the 8-1/8 in mirror aperture and the new bottom edge as exists between the aperture and each original hex flat. It therefore improves base clearance without introducing a uniquely narrow section.

The moving mirror plate uses 3/4 in Baltic-birch sheet. The 1/2 in Baltic-birch stock is shared with the tabletop frame and is used for the cell-adjuster plate, base plate, front pivot plate, and rear rail-support/foot blocks.

10.2 Three adjuster stations

There are three equivalent adjuster stations.

The adjusters are Kozak parts:

- TS250-80-2500 1/4 in-80 adjuster
- TB250-80-625 bushing
- KB250-80 adjustment knob

Use one adjuster, one bushing, and one knob at each of the three stations.

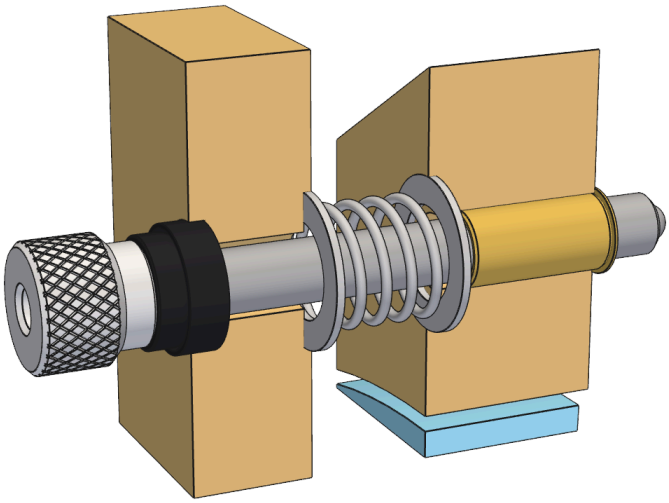


Fig. 12. Adjuster station, plywood and mirror cut away: knob, spherical washer pair, spring and seat washers, bushing, and adjuster screw

Each adjuster station includes a concentric compression spring.

The adjuster bolt passes through the spring and therefore positively retains and laterally locates it. There is no separate spring pocket or spring bore in the plywood.

Adjuster-bushing mounting

Each Kozak TB250-80-625 bushing is bonded directly into the 3/4 in moving mirror plate. No separate metal bushing carrier is used.

Mounting details:

- the bushing's narrow flange is on the mirror-facing / outboard face of the moving mirror plate, opposite the inter-plate spring space; spring/adjuster load therefore tends to seat the flange against the plywood rather than pull it away;
- drill the bushing bore 8.0 mm through the Baltic birch, preferably in a drill press with a sharp brad-point bit and sacrificial backing to minimize breakout;
- the 0.625 in-long bushing barrel is fully contained within the 3/4 in plate and is bonded over its available cylindrical length;
- use WEST SYSTEM G/flex 650 epoxy for the wood-to-metal structural bond;
- bond directly to clean, fresh Baltic-birch bore surfaces; do not CA-harden or pre-seal the bushing bore;
- degrease and lightly abrade the outside of the bushing barrel before bonding without materially reducing its diameter;
- wet both the bore and bushing barrel with a thin continuous epoxy film, while keeping epoxy out of the internal 1/4 in-80 thread;
- seat the flange fully against the outboard face, remove squeeze-out from both faces, and cure with the bushing axis held square to the plate;
- the thin bushing flange is treated primarily as a seating/depth-stop feature. Structural retention is provided by the full-barrel epoxy bond and cylindrical bearing of the bushing in the plywood.

This arrangement adds no hardware thickness to the inter-plate gap.

10.3 Compression springs and seat washers

Components:

- McMaster 2022N186 compression springs
- McMaster 98017A199 stainless spring-seat washers

Each of the three springs is captured between two seat washers, producing:

- 3 springs
- 6 spring-seat washers
- 6 shallow spring-seat-washer counterbores

The counterbores register the washers; they are not spring pockets.

The nominal wood-face-to-wood-face interplate gap is 0.500 in / 12.7 mm at the neutral mirror position.

The spring-seat washer counterbores are:

- diameter: 5/8 in (Forstner cutter);
- nominal depth: 1/32 in / 0.03125 in / 0.794 mm;
- use the same nominal depth for all three seats in the moving mirror plate and all three corresponding seats in the fixed cell-adjuster plate.

The McMaster 98017A199 washers are approximately 0.028–0.036 in thick, so the 1/32 in recess places them approximately flush with the plywood surfaces across their thickness tolerance. With the 0.750 in-free-length McMaster 2022N186 springs and the 0.500 in nominal interplate gap, the springs operate at approximately 0.250 in nominal compression before small washer-thickness and adjustment-position variations.

10.4 Spherical washer interfaces

Each of the three adjuster locations uses:

- McMaster 91131A028 spherical washer set
- one shallow close-fitting plywood socket/counterbore.

Total:

- 3 spherical-washer counterbores

These sockets participate in the mirror-cell gravity load path. The washer OD bears against the counterbore wall, transferring gravity load from the 3/4 in moving mirror plate/adjuster assembly to the fixed 1/2 in cell-adjuster plate.

Target depth:

- approximately 1/16 in / 1.6 mm

The diameter should be a close free fit, not a loose retention pocket. Final cutter diameter should be selected against measured washer OD rather than assuming a nominal oversize woodworking dimension.

Stationary-plate adjuster through-hole

At each adjuster station, use a 1/8 in / 3.175 mm initial alignment guide hole through the fixed 1/2 in cell-adjuster plate. This guide hole establishes a common center for the spring-seat counterbore on the mirror-facing face, the spherical-washer socket on the rear face, and the final adjuster clearance hole.

After machining both counterbores, enlarge the guide hole to a final 5/16 in / 0.3125 in / 7.94 mm through-hole. This is intentionally oversized relative to the 1/4 in Kozak TS250-80-2500 adjuster so the plywood does not become the angular bearing as the moving mirror plate tilts. A 5/16 in hole provides 0.03125 in radial clearance around a 0.250 in adjuster. Across a conservative full 0.500 in plate thickness, a 3° adjuster-axis tilt produces approximately 0.0262 in lateral displacement (0.500 × tan 3°), so the clearance accommodates approximately the full ±3° angular capability of the McMaster 91131A028 spherical-washer interface. The actual effective constraining thickness is somewhat less because of the shallow counterbores.

The machining sequence is: 1/8 in alignment guide hole → shallow counterbores from both faces → 5/16 in final through-hole. The spherical washer, rather than the plywood through-hole, remains the intended angular bearing and load-transfer interface.

CA conditioning

After machining each spherical-washer socket:

1. wick a small amount of McMaster 1818A45 / Loctite 4061 low-viscosity CA into the exposed Baltic-birch fibers;
2. allow it to cure fully;
3. lightly dress the socket back to the required free fit.

The CA hardens the wood; it is not used to glue the washer in place.

Spherical-interface lubrication

Apply a very thin film of light grease only to the mating concave/convex spherical bearing surfaces of each McMaster 91131A028 washer set. Work the pair through its angular range and wipe away excess. This reduces friction and stick-slip as the mirror plate tilts during fine adjustment.

Keep the outside diameter and lower/load-transfer surfaces dry so the lower washer half continues to bear directly against the close-fit CA-hardened plywood counterbore. Keep grease away from plywood, adhesive-bonding areas, and RTV bonding surfaces. The grease is Lucas Oil Products NLGI #2 white lithium grease from shop stock.

10.5 Mirror mounting

The 203 mm mirror is carried within the 3/4 in moving mirror-plate aperture. Its back surface is flush with the rear face of the plate, since both rest on the cure surface (§10.6) during bonding; its front surface is therefore recessed about 1 mm below the front face of the plate.

Six DOWSIL 737 RTV pads are placed radially between:

- the plywood aperture inner surface; and
- the mirror's outer glass edge.

These are compliant radial support/bond pads. Nominal pad geometry is approximately 18 mm circumferential length × 7 mm axial depth × the full approximately 1.7 mm radial bond-line thickness. Place the six pads at 60° intervals. Maintain the approximately 1.7 mm radial gap during cure with temporary non-stick centering shims positioned between pads; do not squeeze the mirror against the plywood or use the RTV pads themselves to force centering. The pad dimensions are practical build targets rather than precision-machined features. No counterbores or pockets are specified for the RTV pads.

Provisional: test adhesion to properly prepared Baltic birch before bonding, to decide whether an adhesion primer is necessary.

10.6 RTV glue-up cure surface

For mirror-plate glue-up and bench-top cure, use a non-stick polyethylene-family release surface beneath the assembly.

- approximately 12 × 12 in UHMW sheet
- large enough for the moving mirror plate (about 279 × 261 mm)
- flat, clean, dry
- no release spray required

This protects the workbench from accidental RTV creep or squeeze-out.

The release sheet is a shop supply rather than part of the finished instrument.

10.7 Structural brackets

Hardware joining the 1/2 in cell-adjuster plate to the 1/2 in base plate:

- 2 × McMaster 8681N11 3 in-long aluminum corner brackets (3 in legs, 1/4 in thick, Ø8.33 mm holes);
- 8 × McMaster 98164A527, 5/16 in-18 × 1-3/8 in 316 stainless button-head hex-drive screws;
- 8 × McMaster 90099A030 5/16 in-18 18-8 stainless heavy-profile nylon-insert locknuts;
- 8 × McMaster 96659A134 5/16 in stainless washers.

Each bracket is flush with a side edge of the plates, with its base leg on the base top extending forward and its upright leg against the front face of the cell-adjuster plate. Each leg has two holes, centered across the leg, 0.75 in and 2.25 in from the outer corner. The screws are pushed through from the bracket side, with the button heads on the bracket's inner faces; the washers and locknuts are on the opposite faces of the plywood, the rear face of the cell-adjuster plate and the underside of the base. Each bracket therefore takes four screws, two through the plate and two through the base, eight in all. The plywood holes match the bracket holes.

The button heads (Ø0.547 in × 0.166 in) are low enough that the heads of the two holes nearest the inside corner clear each other by more than 6 mm, so all four holes are used. The washer and locknut stacks are on the far sides of the plywood and cannot meet.

The 1-3/8 in screw suits the 1/2 in plywood + 1/4 in bracket + washer + 7/16 in locknut stack of about 1.25 in: the thread projects approximately 0.12 in, about 2 threads, beyond the locknut, preserving full locknut engagement without excessive overhang under the base or behind the cell-adjuster plate.

The bracket attachments are through-bolted rather than relying on wood screws. The locknut stacks under the base lie near its side edges, well outside the quick-release screw pattern on the underside (§10.9).

10.8 Mirror safety and transport retention

Front safety retention is intentionally non-stressing:

- loose retainers/fingers prevent escape of the mirror;
- they apply no deliberate optical preload;
- any surface that could touch glass should be soft/non-marring; adhesive-backed F1 felt (McMaster 8334K15) is the provisional facing.

Transport sandwich

For shipping or travel the cell is disassembled and the moving mirror plate, with its mirror and bonded bushings, is held between two covers. The adjusters, springs, and seat washers are removed.

- Each cover is thin hardboard (masonite or similar) cut to the profile of the moving mirror plate, with a clearance hole at each of the three adjuster-bushing centers. There is one cover on the front and one on the rear.
- The mirror's back surface is flush with the rear face of the plate (§10.5), so the rear cover carries a full-face sheet of thin closed-cell foam, which bears evenly on the glass and the plywood.
- The front glass is recessed about 1 mm below the plywood face, and the bushing flanges stand 0.254 mm proud of it. The front cover carries a foam ring that bears on the outer rim of the glass, outside the clear aperture, so it does not touch the reflective surface across the aperture. The ring also acts as a dust gasket. A sheet of optical tissue lies over the mirror face beneath the cover to keep the foam and any dust off the coating.
- Closed-cell foam is preferred, because open-cell foam sheds particles. The foam thickness, about 3 mm, covers the 1 mm recess plus a light compression; it sets the preload and is chosen so the glass is held without stressing it.
- Low-profile thumb screws thread into the adjuster bushings from the front and from the back, with full thread engagement, to clamp the sandwich. The bushing thread is 1/4-80, which common thumb screws do not match. The intended screw is a short Kozak TS250-80 adjuster (available from 5/16 in) with a KB250-80 knob, which threads onto the screw end and can be removed. Lengths of about 5/8 in for the front and 3/4 in for the rear suit a 1/8 in cover and a few millimeters of foam; the engaged lengths from the two sides together must stay below the 15.6 mm barrel length, and the barrel's rear end is 3.2 mm below the rear face of the plate.
- Threaded rods that pass through the bushing bores without engaging their threads, with thumb nuts on both sides of the covers, would also work. They are less preferred: care is needed not to damage the fine bushing threads.
- The covers restrain the mirror axially only; lateral restraint is by the RTV pads. The sandwich is then wrapped in bubble wrap or boxed.

Provisional: the safety retainers are not yet designed. For the transport sandwich: the cover material and thickness (the covers are clamped at 120.65 mm radius, while the foam ring bears at about 100 mm, so the covers must be stiff enough between the clamp points, possibly 1/4 in); the foam material and thickness; the ring's width against the clear aperture; the thumb-screw lengths and their availability (the Kozak catalog pages showed the short screws and the knob as sold out); and the optical tissue.

10.9 Tripod interface

The mirror cell mounts on the tripod through a CAMVATE Manfrotto-type quick-release assembly with sliding plate (Amazon B0B2WBC921) on a Sunwayfoto LB-68R leveling base, whose top mounting stud is 3/8 in-16. The receiver stays on the LB-68R; the removable 501PL/577/504-compatible sliding plate stays attached to the underside of the 1/2 in Baltic-birch mirror-cell base.

The sliding plate is 7.00 in / 177.8 mm long, mounted flush with the front and rear edges of the 7.0 in-deep base, so it spans 0–7.00 in / 0–177.8 mm aft of the front edge and its geometric center lies 3.50 in / 88.9 mm aft of that edge. This is 0.5 in / 12.7 mm aft of the estimated mirror-cell center of gravity (§10.1), which the quick-release system's sliding adjustment can trim.

The plate is attached by four McMaster 92125A196 M4 × 0.7 × 18 mm 90° flat-head screws:

flush flat-head M4 screw → countersunk 1/2 in Baltic-birch base → M4 × 0.7 tapped hole in solid aluminum of the sliding plate.

The screws are in a two-front / two-rear rectangular pattern. Each screw center is 1.875 in / 47.625 mm in from its nearest longitudinal plate end, and the two screws at each longitudinal station are ±0.500 in / ±12.7 mm from the plate centerline. The longitudinal stations are therefore 1.875 in and 5.125 in / 47.625 mm and 130.175 mm aft of the base front edge, with 1.000 in / 25.4 mm transverse spacing within each pair and 3.250 in / 82.55 mm between pairs. These locations lie in solid aluminum, clear of the sliding/clamping bearing surfaces and the central camera-screw slot. The heads finish flush with or slightly below the plywood top surface. The holes are tapped with an M4 × 0.7 tap (3.3 mm tap drill), and the screws are retained with Loctite 243 medium-strength removable threadlocker.

The sliding plate is used without its camera screws. Its camera face, with the slot, goes against the underside of the base. In the base, each screw has a Ø4.5 mm through hole, countersunk 90° to Ø8.4 mm at the top surface, which seats the Ø8 mm head about 0.2 mm below the surface. The 18 mm screw then reaches about 5.5 mm into the plate, roughly eight threads. The base is drilled from the layout below; the plate holes are then marked through the base holes with the plate held against the underside, and tapped.

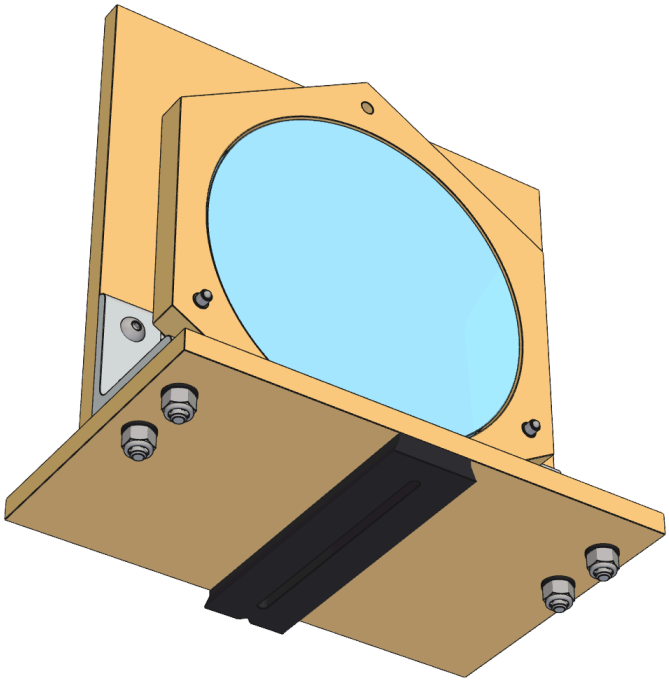


Fig. 13. Mirror cell from below: the quick-release sliding plate on the underside of the base, and the bracket locknuts

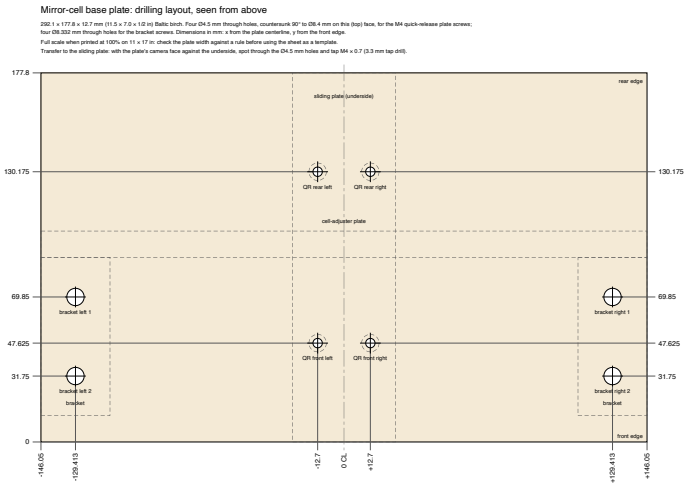


Fig. 14. Base plate drilling layout, seen from above, in cell coordinates (front edge at the bottom)

Provisional: the Ø4.5 mm clearance hole, the countersink diameter, and the depth of the tapped holes in the sliding plate.

11. Provisional areas and next steps

Procurement state and the CAD state of printed parts are tracked in `bom/bom.csv`.

11.1 Provisional areas

Design areas marked **Provisional** in the sections above:

- **Light source** (§6.2, §6.3, §6.7): M3 heatsink-screw length; lead hole size against the lead insulation, cap thread root radius, and Convoy star notch geometry (§6.4); heatsink temperature rise at 700 mA; white-module long-gap focus margin.
- **Cutoff carriage** (§8): clearance to other equipment near the cutoff station.
- **Cassettes** (§8.6, §8.7): filament-cleat geometry.
- **Lens cradle and phone rest** (§9): fit allowances of the rod groove, ball-stop rims, and yokes (including the locating walls on the shoe), the magnet and rod adhesive, and the retaining-band ring size on an assumed modulus (§9.5, §9.8); the phone rest's retainer and safety catch (§9.6); the lens alignment procedure, untested (§9.9). The alignment tolerances rest on assumed lens and camera values, and the contact loads on an assumed lens balance point and phone mass. The inside of the phone case (camera lip height, back and front thickness, openings) is assumed (§9.1).
- **Printed PLA** (§1.1): creep and relaxation of sustained loads, notably the slit-head width-stage preload (§7.7), and heat exposure of printed parts in transport and use.
- **Mirror cell** (§10.5, §10.8): RTV adhesion primer decision; safety retainers; transport sandwich covers, foam, and 1/4-80 thumb screws; quick-release screw hole sizes (§10.9).

11.2 Next steps

1. Trial wire mounting on the cassette filament cleats (§8.6), and check the cutoff carriage's clearance to other equipment on the rail once the layout is set (§8).
2. Fabricate both threaded LED modules: drill and tap the caps and heatsinks, center each board, and set focus on the slit blades; measure the cap thread root radius, and check the lead hole size and the Convoy star notch geometry against the physical parts (§6.4). Confirm that the white module reaches focus; if not, add a thin cap-flange shim (§6.7). Bench-test the heatsink temperature rise.
3. Fit-test the printed yokes on the shoe, the rod groove, and the retaining bands (§9), after measuring the magnet pull on a screw tip, the lens balance point, and the cased phone mass; design the phone rest's retainer and safety catch; test the lens alignment procedure (§9.9).
4. Design the mirror-cell safety retainers and settle the transport sandwich (§10.8), and confirm the quick-release screw hole sizes (§10.9).
5. Assemble the tabletop optical head.
6. Perform a dry optical layout near 3.2 m before making any remaining irreversible holes or bonds.
7. Bond the mirror into the moving plate (§10.5) after a dry fit and an adhesion test.